

The Chemistry of TALSPEAK: A Review of the Science

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ABSTRACT

The TALSPEAK Process (Trivalent Actinide Lanthanide Separation with Phosphorus-reagent Extraction from Aqueous Komplexes) was originally developed at Oak Ridge National Laboratory by B. Weaver and F.A. Kappelmann in the 1960s. It was envisioned initially as an alternative to the TRAMEX process (selective extraction of trivalent actinides by tertiary or quaternary amines over fission product lanthanides from concentrated LiCl solutions). TALSPEAK proposed the selective extraction of trivalent lanthanides away from the actinides, which are retained in the aqueous phase as aminopolycarboxylate complexes. After several decades of research and development, the conventional TALSPEAK process (based on di-(2-ethylhexyl) phosphoric acid (extractant) in 1,4-di-isopropylbenzene (diluent) and a concentrated lactate buffer containing diethylenetriamine-N,N,N',N'',N''-pentaacetic acid (actinide-selective holdback reagent)) has become a widely recognized benchmark for advanced aqueous partitioning of the trivalent 4f/5f elements. TALSPEAK chemistry has also been utilized as an actinide-selective stripping agent (Reverse TALSPEAK) with some notable success. Under ideal conditions, conventional TALSPEAK separates Am^{3+} from Nd^{3+} (the usual limiting pair) with a single-stage separation factor of about 100; both lighter and heavier lanthanides are more completely separated from Am^{3+} . Despite this apparent efficiency, TALSPEAK has not seen enthusiastic adoption for advanced partitioning of nuclear fuels at process scale for two principle reasons: 1) all adaptations of TALSPEAK chemistry to process scale applications require rigid pH control within a narrow range of pH, and 2) phase transfer kinetics are often slower than ideal.

To compensate for these effects, high concentrations of the buffer (0.5-2 M H/Na lactate) are required. Acknowledgement of these complications in TALSPEAK process development has inspired significant research activities dedicated to improving understanding of the basic chemistry that controls TALSPEAK (and related processes based on the application of actinide-selective holdback reagents). In the following report, advances in understanding of the fundamental chemistry of TALSPEAK that have occurred during the past decade will be reviewed and discussed.

Keywords: TALSPEAK, actinide, lanthanide, solvent extraction, HDEHP, lactic acid, DTPA

INTRODUCTION:

Background on the TALSPEAK Process

In Volume 6 of *Ion Exchange and Solvent Extraction* (1974) a review of separation of lanthanides from each other and from actinides was published¹. In that report, Boyd Weaver observed (in describing the TALSPEAK -Trivalent Actinide Lanthanide Separation with Phosphorus-reagent Extraction from Aqueous Komplexes - process) that “*Much fruitless effort was expended searching for extractants to separate the two groups of elements... The separation was only accomplished by considerably modifying the aqueous medium.*” This separation is challenging because 1) the target metal ions exhibit very similar chemistry dominated by the trivalent oxidation state, 2) the radii of the ions decrease by about 20% across the series and the radii of isoelectronic pairs are nearly identical, 3) feeds are typically intensely radioactive, hence radiolytic degradation of reagents (and its effect on extraction conditions) is an issue, and 4) the separation chemistry is based on the very slight increase that is seen in the covalency of interaction between An(III) and Ln(III) cations and ligand donor atoms “softer” than oxygen. The separation is important because ²⁴¹Am dominates the heat load of used nuclear fuel from reactors from about 20-4000 years after the discharge of fuel from a reactor; heat load in the surrounding rock is a primary determinant of the size of a geologic repository for used nuclear fuel or high-level waste from reprocessing.² Americium can be transmuted to less hazardous materials in a reactor, but only after most of the fission product lanthanides have been removed.*

Forty years later, efforts continue to develop industrially-viable approaches to the challenging task of separating trivalent transplutonium actinides from fission product lanthanides. Some significant portion of the new research on this subject focuses on the development of new reagents that exhibit useful selectivity for the (5f) trivalent actinides over the (4f) lanthanides.³⁻⁶ In every case, these approaches focus on taking advantage of the slightly greater covalency that characterizes the bonding of the actinides. The TALSPEAK Process, originally developed by

* Several fission product lanthanides have large neutron capture cross sections thus they compete with Am for fission neutrons decreasing the efficiency of the transmutation process. Several of these lanthanides can be present at higher concentrations than Am, depending on the time since discharge of the fuel from the reactor where it was produced (beta decay of ²⁴¹Pu – representing about 1 kg/ton of fuel, $t_{1/2} = 14.4$ years, at discharge).

Weaver and Kappelmann⁷ at Oak Ridge National Laboratory in the 1960s, remains a viable contender for accomplishing this challenging separation.

A revival of interest in TALSPEAK separations was arguably led by researchers at Argonne National Lab during the course of developing the UREX suite of advanced reprocessing schemes.⁸⁻¹⁰ The UREX process begins with some material control options (based principally on tributyl phosphate extraction from nitric acid media) for managing the bulk of actinides present (U, Np, Pu) using oxidized actinide chemistry. In UREX, the residues (aqueous raffinate) containing all Am, Cm and fission products and reduced concentrations of U, Np, Pu are subjected first to a TRUEX partitioning of Am, Cm, and lanthanides from the remainder of fission and corrosion products and subsequently to a trivalent actinide/lanthanide separation. Though other options were considered in the development of various UREX reprocessing flowsheets, TALSPEAK emerged as the best option for accomplishing this final, chemically challenging separation.

TALSPEAK Process Characteristics. The TALSPEAK process is based on the balance of the relative affinity of cation exchanging solvent extraction reagents and comparably powerful aqueous complexants for the two classes of metal ions. In its simplest operational mode, the extractant exhibits strong affinity for both lanthanides and actinides while the actinide-selective aqueous complexants more strongly binds trivalent actinides. Introducing a “soft-donor” atom (nitrogen and sulfur are most effective) into either the extractant phase or the aqueous phase in principal modifies the natural tendency of trivalent actinides to “follow” trivalent lanthanides to result in the desired selective partitioning of one group from the other. Of course, the task is not quite that simple, as the cationic radii (and hence the electrostatic components of the bonding interactions/solvation) steadily decrease across the respective series. The cationic radius of americium (Am^{3+}), the principal actinide of concern in this context, is nearly identical to that of Nd^{3+} ; this particular separation is often the most difficult to accomplish. In process development, thermodynamic parameters of these ions in hard donor systems are often compared. The actinide isotope of greatest concern is ^{241}Am , which is a low-yield product of neutron capture reactions, but of increasing importance in aged fuel as its concentration increases as its radioactive parent decays (^{241}Pu represents about 0.1% of freshly discharged fuel from LWRs and decays with a 14.4 year half-life to produce ^{241}Am ; the longer-lived ^{241}Am accumulates much more rapidly than

it decays). The lanthanides of interest include La-Dy, some of which are important neutron poisons (Nd, Sm, Pm, Eu, Gd) while others are among the larger contributors to the fission product mass (Ce, Nd, Sm, Y); lanthanides heavier than Dy are not important fission products.¹¹

In the original development of the TALSPEAK process, Weaver and Kappelmann focused on the extractive power of dialkyl phosphoric and phosphonic acid extractants balanced against the actinide selectivity of aminopolycarboxylic acid complexing agents. Bis-2-ethylhexyl phosphoric acid (HDEHP), which ultimately emerged as the favored extractant, can extract trivalent f-elements from dilute (0.1-0.01 M) mineral acid solutions; aminopolycarboxylic acid complexants generally require somewhat higher pH for both adequate solubility and appropriate complex strength. Of necessity, TALSPEAK evolved toward buffered aqueous media in the pH 2.5-4.5 range. The cation exchanging extractant exhibits a moderate sensitivity to the decreasing cationic radius of lanthanide and actinide cations that occurs with increasing atomic number.

Thermodynamic extraction efficiency increasing on average by about 2 kJ/mol for each successive member of the series. The extractant typically does not distinguish between Ln^{3+} and An^{3+} ions of comparable size. The aminopolycarboxylate ligands tend to complex trivalent actinides more strongly than the lanthanides of similar ionic size, though complex stability patterns tend to exhibit different trends based on complex stoichiometry and ligand denticity (which will be considered in more detail below).

To solve issues of solubility, to accommodate the need for pH control, and to improve phase transfer kinetics, Weaver and Kappelmann considered a variety of carboxylic acid buffers. The most efficient pH range for operation of various TALSPEAK options lies between pH 3.0 and 4.5. Establishing acceptable phase transfer kinetics and pH control ultimately demanded that the concentration of the buffers be maintained in the range of 0.5 to 2.0 M. Ultimately, Weaver and Kappelmann settled on a combination of di-(2-ethylhexyl)phosphoric acid (HDEHP) in 1,4-diisopropylbenzene balanced against a concentrated sodium lactate buffer containing an appropriate concentration of diethylenetriamine-N,N,N',N'',N''-pentaacetic acid (DTPA) operating in the range of pH 3.0-4.0 as a baseline formulation for TALSPEAK separations. They observed that aliphatic diluents gave higher extraction efficiency, but reduced actinide/lanthanide selectivity, hence the hybrid diluent. Over the past four decades this chemistry has been investigated periodically with a variety of different concentrations of each component, with

alternative cation exchanging extractants, substituted into other separation platforms, even adapted to processes attempting total spent fuel management strategies based on a single extraction platform (e.g., CTH^{12,13} and DIDPA¹⁴ processes). Actinide-selective aminopolycarboxylate complexants in buffered media have also been adapted to separation platforms not based on liquid cation exchanging extractants with mixed success.¹⁵

Though a number of “fundamental” studies have been conducted on some features of TALSPEAK during the years since its inception (radiolysis, phase transfer kinetics, extraction reactions via stoichiometric slope analysis), more studies have focused on application development and optimization. The fundamental chemistry of TALSPEAK was reviewed in detail in 2007, coinciding with its integration into UREX process development.¹⁶ In that report an attempt was made to correlate numerous independent observations to develop a coherent picture of the potential gaps in understanding of the TALSPEAK process. As usual, correlating the observations of numerous investigators (or teams of investigators) operating with various constraints in widely dispersed locations provides some insights, but only a limited number of clear consistencies. Focusing in principal on the baseline TALSPEAK process (0.1-0.5 M HDEHP in NPH or DIPB, 0.5-2.0 M Na/H-Lactate, 0.05-0.10 M DTPA, pH 3.0-4.0), three observations have been made concerning issues to be overcome prior to process-level application:

- 1) Extraction of lanthanides and actinides decreases significantly as pH rises, hence pH control is essential for process-scale applications,
- 2) Extraction kinetics are often slow enough to require longer than desirable phase contact times,
- 3) Process-worthy system performance is closely tied to a relatively narrow band of conditions, in particular pH.

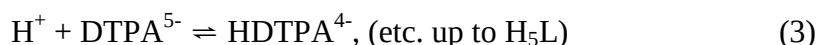
The Thermodynamics of TALSPEAK. Between 1960 and 1980, a considerable amount of thermodynamic data describing the interactions of lanthanides and actinides with primary components of conventional TALSPEAK was reported in the open literature.¹⁷ Numerous studies reported stability constant values for lanthanide carboxylate and aminopolycarboxylate

complexes, a smaller number of equilibrium constants for trivalent actinide complexes, and a few studies reported reliable trans-lanthanide extraction data for HDEHP extraction systems. These data enabled (with a few approximations) the calculation of predicted solvent extraction partitioning trends. Because there is no standard “reference state” (ionic strength, temperature, ionic medium) in such studies, one challenge of predicting the thermodynamics of TALSPEAK process performance is the requirement that the relevant thermodynamic data must be compiled and compared with a cautious eye toward internal consistency.

Of particular interest in thermodynamic modeling of TALSPEAK was the relationship between lanthanide extraction patterns from concentrated lactate buffer solutions containing DTPA as a function of pH. Though the basic phase transfer reaction has a clear inverse third power dependence on $[H^+]$,



the acid dependence of the competing aqueous metal complexes with $H_{5-n}DTPA^{n-}$ and the corresponding lactate complexes is more complex. The aqueous equilibria of importance include that of the phase transfer reaction (equation 1), protonation of lactic acid and DTPA (equations 2 and 3),



The formation of 1:1, 1:2, 1:3 lanthanide and actinide lactate complexes,



and complexes of Ln^{3+} and An^{3+} with DTPA to form $M(DTPA)^{2-}$ and $M(H(DTPA))^-$ complexes,



In the previous report, speciation diagrams were prepared for each of the lanthanides, yttrium and Am^{3+} to assess differences in speciation. It was clear in this exercise that the complexometric balance between HDEHP and DTPA was primarily responsible for the observed patterns of extraction and group separation, though lactate complexes gained importance in its potential impact on extraction efficiency at higher lactate concentration for the lightest and heaviest lanthanide ions.

Basic thermodynamic modeling calculations reported in reference 18 (based on the assumption that the above equations are the only reactions to consider) indicate that in the range of pH 3.0-4.5 the extraction reactions of Ln^{3+} and An^{3+} should be nearly constant, with a slight increase in distribution ratios at pH > 3.2 (Figure 1[†]). The most important DTPA-HDEHP equilibrium at the normal pH = 3.5 operational condition is:



For trivalent lanthanides the equilibrium lies to the product side, for trivalent actinides to the reactant side.

In contrast with this prediction of near acid independence, numerous reports in the literature on conventional TALSPEAK (and related systems) experimentally demonstrate decreasing extraction (in some reports, relatively steep changes) as pH increases.^{7,19,20} Though the thermodynamic modeling calculations do not predict the pH profile observed in TALSPEAK applications, they do accurately predicted the trans-lanthanide pattern of extraction and the Am/Ln separation efficiency that is observed at the baseline TALSPEAK pH of 3.5 (Figure 1b).[‡] Coupled with the general lack of a chemical explanation for the slow phase transfer kinetics, these results motivated a series of studies directly probing the chemistry of TALSPEAK.

FIGURE 1 NEAR HERE

[†] The reader is referred to reference 16 for the details of this calculation.

[‡] The validity of such calculations is highly dependent on stringent adherence to developing an internally consistent set of thermodynamic parameters.

Summary of 2007 review: TALSPEAK process profile

The combined review of the prior literature and computational analysis of the thermodynamics of TALSPEAK described in the 2007 review clearly describe a system based on complex chemical interactions. Slope analysis results from the previous literature hint at, but ultimately provide unsatisfactory explanations for, many of the observations that have been made during efforts to develop viable TALSPEAK processes. Aside from the issues of slow phase transfer kinetics and unexpected/undesirable acid dependence in the extraction reactions, the earlier review describes selected overarching features of TALSPEAK separations that are reasonably well known, as follows:

- 1) The primary roles of the components (lactate = buffer, DTPA = actinide holdback reagent, HDEHP = lanthanide extractant) do not account for the relative impact of each species on process performance (only DTPA is consistently predictable).
- 2) High concentrations of lactate buffer increase the solubility of DTPA²¹
- 3) Lactate serves as a radiation scavenger and increases the effective lifetime of DTPA in the process
- 4) The order of extraction of trivalent lanthanides and Am³⁺ is Am << Nd < Pm < Sm = Eu < Pr < Ce = Gd < La < Tb < Dy < Ho < Er < Tm < Yb < Lu. This pattern is consistent with predictions based on the relative stability of lanthanide and Am complexes with HDEHP vs. DTPA; thermodynamic data for lanthanide and actinide complexation by DTPA, lactate, and HDEHP predict relatively closely the observed (pseudo-parabolic) pattern of relative extraction efficiency and separation.
- 5) Lactate effectively serves as a phase transfer catalyst
- 6) An option for a citrate-based TALSPEAK has been described in the prior literature;²² this option may offer advantages in the management of fission product zirconium
- 7) At high concentrations, lactate is reportedly responsible for phase separation issues (increased dissolution of HDEHP in the aqueous media) while at lower concentrations it assists with phase disengagement.⁷

- 8) Lactic acid impacts the kinetics of the complexation between DTPA and metal ions in the aqueous phase, possibly forming a mixed complex, probably as a transient species.²³
- 9) Several reports have suggested (based on slope analysis primarily) the presence of ternary complexes ($M(lac)(HDEHP)_x$ and $M(lac)(DTPA)$) represent an unknown aspect of TALSPEAK, as there are no fundamental data establishing their existence, only inferences of the effect of lactate on cation partitioning.
- 10) The complexes of lanthanide cations by polyaminopolycarboxylic acids are known to be kinetically hindered, with metal complex dissociation reactions proceeding via a sometimes involved series of steps^{24,25}, though in buffered aqueous media the kinetics of these reactions are too rapid to account for the observed slow phase transfer kinetics of TALSPEAK
- 11) The phase transfer mechanistic studies done previously by Kolarik and Kuhn²³ and by Danesi and Cianetti²⁶ provide a workable and useful description of the operation of the process, but do not establish a clear molecular-scale explanation for the observed kinetics or an explanation of why lactate is (or is not) a unique phase transfer catalyst; each study cites interfacial decomplexation reactions as rate limiting processes.
- 12) Analysis of the thermodynamics of extraction based on literature data indicate that the observed Ln/An extraction pattern results from a counterbalance between a highly effective solvent extraction reagent and a powerful aqueous complexant. As will be seen below, matching weaker extractants with weaker aqueous complexants can yield similar system performance.

Because of the gaps that exist in the fundamental understanding of the chemistry of TALSPEAK, attempts to adapt the basic philosophy of TALSPEAK to other separations systems will have to rely to a large degree on trial and error approaches. This potentially powerful lanthanide-actinide partitioning technique could have far greater impact on spent fuel partitioning efficiency if additional studies of its basic features were conducted. This analysis of the known literature inspired a number of detailed investigations of the molecular scale interactions that drive, limit and generally complicate the basic chemistry of TALSPEAK.

Two specific features of this analysis of TALSPEAK emerged as areas of particular interest for further consideration 1) the clear mismatch between the predictions of slight pH dependence of the extraction as compared with the repeated reports of a strong negative correlation between increasing pH and extraction efficiency;¹⁸ the observed disagreement between thermodynamic model predictions and observed behavior can be most simply considered to result from phenomena that cause a) extraction strength to be lost or, b) holdback strength to be gained, and 2) the absence of a clear picture of the relationship between the chemical kinetics of the reactions that must define phase transfer reactions and the observed complexity of the phase transfer kinetics.

- On the acid dependence question, the prospect of involvement of ternary complexes in either the organic or aqueous phases (or both) offered an obvious target for investigation and has been previously implicated.^{7,19,20} The earlier reports of reduced extractant dependence in TALSPEAK extraction reactions, explained in the prior literature as a result of extraction of $\text{Ln}(\text{Lac})^{2+}$ species, was the most widely considered explanation. As this conclusion was based principally on the results of slope analysis experiments, it was thought that a more direct probing of this possibility was in order. The other possible explanation for declining extraction of metal ions that is seen as the pH increases is the possible loss of extraction efficiency (in contrast to the predicted slightly rising extraction predicted by the HDEHP-DTPA modeling) that could result from a change in the extractant organization or another competing reaction that reduces the availability of HDEHP for the essential task of lanthanide extraction.
- On the question of the kinetics of the phase transfer reaction (and the evident catalysis of lanthanide phase transfer by lactate), the complementary questions of ligand exchange reactions in the aqueous and organic phases separately were an obvious complement to the results of Kolarik and Danesi (which concluded that hindrance was arising from interfacial species, implicitly intermediate between the thermodynamically stable species of $\text{Ln}(\text{DTPA})^{2-}$ and $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$) offered another intriguing target for investigation.

Scope of this Report

In the following, the results of experimental investigations of the fundamental basis of the conventional TALSPEAK process that have been completed since the 2007 review are described and an attempt is made at integration of these observations. The reported work includes examination of the thermodynamics and kinetics of the basic chemistry of the system. It also examines metal ion and complexant speciation and the effect of supramolecular organization on extraction reactions. Most components have been investigated using a combination of radiometric analysis methods and direct spectroscopic interrogation of TALSPEAK molecules and ions (or representative model species). In the closing sections reports on minor adjustments of TALSPEAK in process development, investigations of process monitoring and control relevant to TALSPEAK, and updated approaches to application development of TALSPEAK process alternatives are discussed.

Thermodynamic Modeling of TALSPEAK

As there were several reports of a strong acid dependence in TALSPEAK extraction and since the thermodynamic modeling exercise in the earlier review did not correlate with observed reports from the separations literature, it was deemed prudent to conduct a carefully controlled investigation to examine the acid dependence for extraction of all lanthanides and Am^{3+} . Trans-lanthanide studies are considered to be integral to the discovery process because La-Dy and Y are important fission products, but also because trans-lanthanide studies can provide useful insights of broad impact. The system performance experiments focused on 1 M lactate/0.05 M DTPA, pH 3.6 solutions in which lanthanides and Am are extracted into 0.2 M HDEHP/1,4-DIPB solutions; all experiments were completed under carefully controlled conditions of constant ionic strength[§], pH, and temperature.¹⁸ The prior reports from the solvent extraction literature focused primarily on process optimization rather than maintaining conditions designed to support thermodynamic modeling of the system. It was considered that perhaps the mismatch between observed acid dependence and model predictions might arise from the inconstant activity of concentrated

[§] It is probably not necessary to maintain constant ionic strength in TALSPEAK process operation, but to facilitate understanding of the system control it is a desirable system constraint, particularly for model predictions.

lactate buffer media. By controlling ionic strength and temperature, the impact of this possibility could be reduced.

It was reported by Nilsson and Nash¹⁸ that even under such carefully controlled conditions extraction of both lanthanides and Am^{3+} decreased with increasing pH and also with increasing temperature (at constant pH 3.6 and fixed ionic strength, 2.0 M, Figure 1a, above). As pH increased and distribution ratios dropped, $\text{Ln}^{3+}/\text{Am}^{3+}$ separation factors remain approximately constant, within the accuracy of the measurements; of course in reality the utility of the “separation” decreased as lanthanide extraction decreased with increasing pH. Adopting the best-available compilation of thermodynamic data from the prior literature,¹⁷ cation partitioning calculations failed to predict the experimentally observed loss of extraction as pH increases. Calculations again indicated that for the simplest model dominated by DTPA complexes and HDEHP extracted complexes a rise in pH should produce a slight increase in extraction of both lanthanides and actinides. This modeling exercise did predict the observed lanthanide and Am pattern of extraction (Figure 1b).

To assess whether the thermodynamic data being used in the modeling was appropriate to the task, Nilsson and Nash¹⁸ examined also the temperature dependence of TALSPEAK extraction reactions. Completing a series of lanthanide and Am extraction reactions as a function of temperature enabled the experimental determination of a conditional extraction enthalpy for the net extraction reaction (pH 3.5), as shown in equation 7. Conditional enthalpies for lanthanide extraction under TALSPEAK conditions that were calculated from the adjusted slopes of $\ln D$ versus inverse temperature ($1/T$, K^{-1}) are shown in Figure 2. The two sets of data using constant nitrate (designated run 4) or constant sodium ion concentrations (run 5) are shown separately in the figure. Also shown in the figure are enthalpy values for the extraction of lanthanides from mineral acid solutions by HDEHP in *n*-heptane determined by the Van’t Hoff method,^{27, 28} and by biphasic titration calorimetry of lanthanide extraction by HDEHP in *n*-dodecane.²⁹ Except for Am, the extraction reactions are exothermic, for both the HDEHP extraction of these metal ions from mineral acid solutions, and from TALSPEAK aqueous media. In each lanthanide system the least exothermic reaction is seen for Nd^{3+} . Most surprising is the parallel patterns that are seen for the TALSPEAK and mineral acid solution systems. For the lightest lanthanides enthalpy values overlap within the uncertainty of the measurement. For the lanthanides heavier than Nd, the

extraction from TALSPEAK aqueous phases is 5-10 kJ/mol less exothermic. Unfortunately, there are too many potential variables in the composition of these media to allow a more detailed analysis/attribution of potential cause. This observation is further addressed with the results of new thermodynamic data, as discussed below.

FIGURE 2 HEAR HERE

Free Energies of ligand protonation and complexation reactions

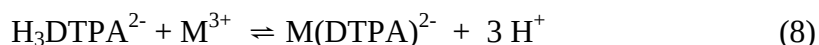
One final limitation to the accuracy of the thermodynamic modeling of TALSPEAK was the reliance on Ln-DTPA stability constant and ligand protonation constants from the prior literature. In the calculations described above, it proved impossible to find in the existing literature a completely consistent set of thermodynamic equilibrium constants describing all possible reactions. The most important matching of conditions was made between ligand protonation and complexation constants for lactate and DTPA. Literature data for extraction equilibria are far less consistent (and suffer the effects of a wider array of variables). Of particular concern was the need to match $I = 2.0$ M lactate data with $I = 0.1$ M DTPA data. To reduce the uncertainty of the modeling from this factor, fresh estimates were made of the thermodynamic parameters describing lanthanide-DTPA interactions at 2 M ionic strength (NaClO_4) and 25.0°C .³⁰⁻³²

Protonation equilibrium constant data reported by Grimes are compared with earlier reports of similar data at lower ionic strength in Table 1. Lanthanide complexation equilibrium constant values are given in Table 2. DTPA protonation constants are defined for the successive protonation equilibria noted schematically in equation 3. Lanthanide-DTPA complexation equilibrium constants are defined as in equations 5 and 6. The protonation constants most affected by ionic strength are the first and second equilibria. From examination of the 0.1 M and 2.0 M ionic strength data, it is apparent that, while selected equilibrium constants are larger at 0.1 M than at 2.0 M, the relative magnitude of protonation and metal complex stability vary consistently, thus either set might be expected to produce a reasonable representation of equilibrium speciation in TALSPEAK systems. Care must be taken in assembling data sets for such modeling to match as closely as possible the conditions for parameter sets to maximize accuracy of predictions.

TABLE 1 NEAR HERE

TABLE 2 NEAR HERE

It can be seen in an examination of these data that equilibrium constants are 50-100 times lower at 2.0 M ionic strength than at 0.1 M (as is typically seen). However, ligand protonation equilibrium constants are also lower at 2.0 M ionic strength (though the first and second ligand protonation reactions are more significantly impacted). Considering that the dominant DTPA free ligand species at pH 3.6 is $\text{H}_3\text{DTPA}^{2-}$, the effective metal complexation reaction in TALSPEAK aqueous phases is:



for which a few representative equilibrium constant values ($\log \beta_1 - \log (K_{h1} \cdot K_{h2} \cdot K_{h3})$) are: $\text{La}^{3+} = -3.88$ (I = 0.1 M), -4.27 (I = 2.0 M), $\text{Nd}^{3+} = -1.76$ (I = 0.1 M), -1.96 (I = 2.0 M), $\text{Eu}^{3+} = -0.97$ (I = 0.1 M), -1.16 (I = 2.0 M), $\text{Tb}^{3+} = -0.65$ (I = 0.1 M), -1.04 (I = 2.0 M). It can be seen that, although the metal complex “stability constants” may differ significantly, the compensating change in affinity for H^+ tends to offset the impact of this change. A more important impact would be seen in comparing the relative strength of competing complexants (e.g., lactate vs. DTPA) if the equilibrium constants were not determined at the same ionic strength and temperature.

It is also necessary in some cases to consider the internal consistency of tabulated thermodynamic parameters. Among the compiled data¹⁷ describing lanthanide lactate complexes at 2.0 M ionic strength, as shown in Figure 3, most of the reported data is internally consistent. However, selected values in the compilation are not consistent with the other 80% of the data (they have been eliminated from this plot). For purposes of the modeling to be shown below, polynomial functions were developed (Origin 7.5) and the suspect data were “improved” with estimates based on interpolation (a reasonable approach to refining thermodynamic data trends probably most reliably employed only within the lanthanide series) shown in Table 3. Using this internally-consistent data set, speciation plots were constructed.

FIGURE 3 NEAR HERE

TABLE 3 NEAR HERE

Using the data analysis and plotting routines of HySS 2009, aqueous phase speciation plots (extraction equilibria not considered) have been developed for the lanthanides for a TALSPEAK-relevant aqueous medium comprised of 0.01 M DTPA, 2.0 M Na/HLac, I = 2.0 M, T = 25°C in the TALSPEAK operational pH range of 1-4 (figure 4) where speciation plots are shown for La^{3+} to Gd^{3+} and Yb^{3+} . It can be seen that below pH 2 lanthanide lactate complexes dominate aqueous speciation. At the pH 4 upper limit of the plots LnDTPA^{2-} is the dominant species, though only just so for La^{3+} ; the increasing relative strength of lanthanide DTPA complexes from La^{3+} to Nd^{3+} actually account for the decline in extraction that is observed for these lanthanides in TALSPEAK. In the pH 2-4 region, the complexes $\text{Ln}(\text{Lac})_n^{3-n}$ and LnDTPA^{2-} are competitive (within the constraints of the concentration limits selected). Based on the pH at which the LnDTPA^{2-} and $\text{Ln}(\text{Lac})_3$ curves intersect, it is obvious that the competitive balance shifts increasingly toward $\text{Ln}(\text{DTPA})^{2-}$ from La^{3+} to Gd^{3+} . The reverse shift toward a more competitive balance between 0.01 M DTPA and 2.0 M lactate seen in the Yb^{3+} curve is representative of the trend seen in the second half of the series, in which the DTPA holdback effect becomes less competitive with HDEHP extraction and the lanthanides heavier than Nd are successively extracted more strongly. This pattern arises because lanthanide lactate complex stability continues to increase across the series, while $\text{Ln}(\text{DTPA})^{2-}$ stability levels off for the ions heavier than Gd^{3+} .

FIGURE 4 NEAR HERE

Speciation calculations assessing the relative stability of the $\text{Eu}\cdot\text{DTPA}^{2-}$, $\text{Eu}(\text{HDTPA})^-$, and free DTPA ligand in 0.1 M KNO_3 , 2.0 M NaClO_4 and 2.0 M NaO_3SCF_3 indicate a somewhat diminished importance for the $\text{Ln}(\text{HDTPA})^-$ complexes at 2 M ionic strength than at 0.1 M. Under TALSPEAK relevant conditions, it is seen that Ln-lactate and Ln DTPA complexes are competitive for the lightest and heaviest lanthanides (in the aqueous speciation), but in the middle of the series $\text{Ln}(\text{DTPA})^{2-}$ is the dominant aqueous species.

FIGURE 5 NEAR HERE

Complexation Enthalpies

In a titration microcalorimetry investigation, Grimes³² reported results that could only be interpreted to indicate that Eu DTPA complexes were weaker in 1.9 M lactate media than in 0.2 M lactate media (at constant 2.0 M ionic strength and temperature). This unexpected result points to a specific ion activity effect related to lactate solutions. Specific ion effects of this sort can compromise the accuracy of thermodynamic modeling predictions in aqueous media as complex as TALSPEAK aqueous phases. Furthermore, the balance between enthalpy and entropy in complexation reactions can provide useful insights into cation binding trends. To address this observation and to gain insights into the limits of titration calorimetry, experiments have been completed examining the thermochemistry of Ln-DTPA complex formation reactions.³³

The ΔH values for complexation of the rare earth metals by DTPA in 2.0 M NaNO_3 at 25.0°C and corrected for enthalpies of protonation are shown in Table 4. They are compared to results obtained previously^{39,40} at lower ionic strength. At either ionic strength, the complexation enthalpies of all lanthanides according to equation 6 (above) are exothermic. The ΔH values increase in magnitude from lanthanum through europium, level off through dysprosium, and decrease thereafter such that ΔH_{Lu} is similar to ΔH_{La} . The reaction becomes approximately 8-9 kJ/mol less exothermic as the medium is shifted toward infinite dilution at 0.1 M ionic strength. The difference in magnitude of ΔH_c between the middle lanthanides and those towards the ends of the series is more pronounced in NaClO_4 media than in the NaNO_3 media.

TABLE 4 NEAR HERE

The ΔG_c values determined by potentiometric titration were combined with the ΔH_c values determined in the calorimetric experiments to calculate ΔS_c values, shown in Table 5. At 0.1 M and 2.0 M ionic strengths, a largely linear increase in ΔS is seen across the series. These are the same conditions under which the largest mid-series decrease was observed in ΔH for this reaction. As with ΔG and ΔH , dependence of ΔS on ionic strength at high supporting electrolyte concentrations is minimal, but ionic strength has an appreciable effect as supporting electrolyte concentration is lowered toward infinite dilution.

TABLE 5 NEAR HERE

The pattern of increasing ΔG_{LnDTPA} from La to Eu in this series is revealed by the information in Figure 6 to arise from a changing complexation enthalpy while entropy remains largely constant. From Eu-Dy both entropy and enthalpy remain nearly constant (thus the free energy is also constant). From Dy to Lu complexation enthalpies become steadily less exothermic while a compensating entropy increase compensates to maintain a stable free energy. It is not unreasonable to speculate that beyond Dy the decreasing size of the lanthanide ion combined with the comparative bulk of DTPA lead to reduced space for residual hydration of $\text{Ln}(\text{DTPA})^{2-}$ and a resultant increase in entropy resulting from the increased release of hydration sphere water molecules to the bulk.

FIGURE 6 NEAR HERE

The determination of lanthanide-DTPA complexation enthalpies at high ionic strength enabled a comparison of the observed enthalpies of the individual components of TALSPEAK with the observed net enthalpy of the process as determined by Nilsson¹⁸ (Figure 2). To do this, the individual enthalpies of each step in the series of reactions required to move a lanthanide ion from the aqueous phase to the organic phase were summed. This model starts with the lanthanide ion in a stable complex with DTPA in the aqueous phase. It is necessary (thermodynamically) to first de-complex the lanthanide from DTPA, then extract the lanthanide into the organic phase (releasing three protons per extracted metal ion), and finally to re-protonate DTPA^{5-} to $\text{H}_3\text{DTPA}^{2-}$.^{**} The observed enthalpies for the DTPA decomplexation reaction, the HDEHP-facilitated phase-transfer reaction, and all relevant protonation and deprotonation reactions were summed to obtain the heat of extraction one would expect in an ideal TALSPEAK system. The data used to construct the model for europium are provided as an example in Table 6. The ideal net reaction enthalpy was then compared to the experimentally determined enthalpy of the TALSPEAK process, in the case of Eu^{3+} showing a discrepancy of 27.7 kJ/mol for the europium extraction.

^{**} The possibility that the liberated protons might be absorbed by the lactate buffer instead of DTPA^{5-} was considered. However, DTPA^{5-} and HDTPA^{4-} are both significantly more basic than Lac^- . The formation of HLac could be expected to compete with the formation of $\text{H}_3\text{DTPA}^{2-}$, but both protonation reactions have similar enthalpy values; which reaction predominates probably does not significantly affect the model.

TABLE 6 NEAR HERE

As shown in Figure 7, the energetic discrepancy observed for the extraction of europium in an ideal system vs. an experimental system remains relatively constant across the entire lanthanide series. The similarity between the experimental enthalpies of the extraction reaction and the full system has previously been taken to indicate that the extraction step is the primary contributor to the heat evolved by the TALSPEAK process and that the DTPA decomplexation step is only a minor contributor.¹³ In light of the thermodynamic data produced by this study, it seems that the breakup of the MDTPA²⁻ complex must have a large effect on the overall enthalpy of a TALSPEAK extraction. This enthalpy contribution is apparently roughly compensated for by endothermic contributions from some unknown reaction or set of reactions. These reactions are likely related to the poorly-understood behavior of the lactic acid/lactate buffer used in conventional TALSPEAK. It is quite conceivable that the missing heat represents a contribution from an endothermic heat of dilution unique to the solvent extraction system that was not present in the calorimetry experiments.

FIGURE 7 NEAR HERE

Are Ternary Complexes Important?

As all manner of thermodynamic modeling based on known equilibria proved to offer no explanation for the observed acid dependence of TALSPEAK and the prior TALSPEAK literature repeatedly invoked mixed ligand complexes to explain selected features of TALSPEAK, it was rational to examine the possibility of mixed-ligand complexes in conventional TALSPEAK. Jensen and co-workers⁴¹ conducted spectrophotometric investigations of Nd³⁺-DTPA-lactate media complemented by Eu³⁺ fluorescence studies and thermometric titrations. These authors report no evidence via any of these techniques for addition of lactate to LnDTPA²⁻ complexes even at 0.75 M total lactate. They conclude that, if such species are present in TALSPEAK aqueous media, they are too rare to account for the consistently observed pH dependence of the system. As will be seen below, it is the full occupation of lanthanide “coordination sites” by DTPA that inhibits the formation of ternary complexes in TALSPEAK

aqueous phases. Substitution of aminopolycarboxylic acid complexants of lower denticity does permit the formation of ternary complexes (though such species are still not implicated in the strong and unexpected pH dependence of TALSPEAK). It will also be demonstrated below that mixed $\text{Ln}(\text{Lac})(\text{HDEHP})$ complexes are important in the organic phase, but apparently only under conditions of high metal loading. Ternary complexes are also implicated as transient species on TALSPEAK phase transfer and aqueous phase complexation reactions, as will be discussed below.

Lactate Partitioning/Water Miscibility

Finding no enlightenment in the basic thermodynamics of the TALSPEAK system, the search for an explanation for the TALSPEAK pH dependence shifted toward the lactate buffer system. Utilizing ^{14}C -labeled sodium lactate, the partitioning of lactate between TALSPEAK aqueous and organic phases was probed directly by Grimes.⁴² In the prior literature on TALSPEAK, organic phase lactate was implicated as an explanation for reduced extractant dependence in selected TALSPEAK studies.¹⁹ Weaver and Kappelmann indicated that partitioning of HDEHP into the aqueous phase could be induced at high lactate concentrations; certainly, one possible explanation for declining lanthanide extraction efficiency with increasing pH could correlate with the loss of extractant, either to the aqueous phase or by some other means. The availability of labeled lactate and facilities to handle it offered a clear pathway to directly probing this facet of TALSPEAK chemistry.

Grimes⁴² found that, contrary to information in the prior literature in which lactate partitioning was correlated with lanthanide extraction, lactic acid partitions into the TALSPEAK organic phase independently of the amount and identity of the metal ions that are present. In Figure 8, it can be seen that in 0.001 M solutions of Y^{3+} , La^{3+} , Nd^{3+} , or Eu^{3+} the lanthanide ions are quantitatively (or nearly so) extracted by 0.2 M HDEHP from all lactate media between 0.01 and 2.0 M at pH 3.6. In the low lactate limit, the lanthanide ions are extracted independent of lactate, no doubt via the formation of $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$ complexes. In the low lactate limit, $[\text{Ln}]_{\text{org}}$ is 50-100 times greater than $[\text{HLac}]_{\text{org}}$; in the high lactate limit $[\text{HLac}]$ is 40-80 times higher than the extracted lanthanides. From these results, it is clear that the predominant lanthanide extraction mode is not via $\text{Ln}(\text{Lac})(\text{DEHP}\cdot\text{HDEHP})_2$, as suggested in several prior references; it is equally clear that lactic acid (or possibly sodium lactate) is extracted independently of

lanthanide extraction. A reasonable conclusion that might be drawn is that in a system this complex, there are likely multiple competing pathways for extraction reactions to occur.

FIGURE 8 NEAR HERE

To probe lactate extraction further, these authors examined lactate partitioning as a function of pH (Figure 9a and with different concentrations of Na^+ present in the aqueous phase (Figure 9b). The results in Figure 9a demonstrate the unusual and unexpected result that lactate extraction actually increases as the pH increases from pH 3.0 to 4.2; assuming that there are no important specific ion effects and that the pK_a value for lactic acid is constant at 3.6, the aqueous phase species distribution is 25% Lac^- /75% HLac at pH 3.0, 50% each at pH 3.6, 75% Lac^- /25%HLac at pH 4.2. Though the distribution ratios are not very high, they do indicate a significant salting out effect arising from the increasing concentration of lactate ion. The results shown in figure 9b demonstrate that lactate extraction and that of Na^+ are not necessarily correlated, implying that the primary extracted lactate species is likely to be HLac.

FIGURE 9 HERE

The partitioning of lactic acid as a function of increasing concentrations of HDEHP/1,4-DIPB is demonstrated in Figure 10. Lactic acid distribution ratios (for partitioning from 1.0 and 2.0 M pH 3.6 lactate solutions) into HDEHP/1,4-DIPB media increase in approximate proportion to $[\text{HDEHP}]^2$ reaching maximum values of about 0.2 in 1 M HDEHP; in this limit the apparent concentrations of lactic acid in the organic phase is 0.17 M for the sample in contact with 1 M lactic acid buffer, double that for the 2 M lactic acid system. If the apparent limiting slope of 2 is used to define the organic phase lactate species as $(\text{HLac} \cdot (\text{HDEHP})_2)$, the effective concentration of free extractant available for lanthanide extraction (at 1.0 M HDEHP) is reduced to $(1.0 - 2 \cdot 0.34) = 0.32$ M (monomer equivalent) in contact with 2 M lactate. If it is assumed that the hypothetical $\text{HLac}(\text{HDEHP})_2$ species does not participate in lanthanide extraction reactions and that the lanthanide extraction equilibrium is represented by equation 1 (based on the dimer concentration), the concentration of the extractant $((\text{HDEHP})_2)$ available for lanthanide extraction is reduced to 0.16 M. With the 1:3 ($\text{Ln}:(\text{HDEHP})_2$) stoichiometry taken into account, the potential suppression of lanthanide extraction is equivalent to approximately 250. Considering that HLac extraction increases with increasing pH, the competing extraction of HLac by HDEHP represents a viable

explanation for the unexpected pH dependence of lanthanide extraction in conventional TALSPEAK. Given the evident complexity of the overall interactions, model development to account for all relevant reactions in this system appears to represent a monumental challenge for solution chemistry.

FIGURE 10 NEAR HERE

Finally, Grimes³² reports also the results of studies of the partitioning of water and lactic acid into TALSPEAK organic phases as a function of lactate concentration and [HDEHP] (Figure 11). It is clearly seen that lactic acid extraction increases with [HDEHP]. None of these systems give any indication of a lactic acid saturation effect at the highest concentrations of lactate, implying that additional extraction of lactic acid (without achieving saturation) is possible. The water content of the extractant phase increases with [HDEHP], at the lower lactate concentrations remaining in the range of 10-20% H₂O to HDEHP. For samples equilibrated with 2 M lactate (pH 3.6) water content increases by a factor of 10; for the 1 M HDEHP/2 M lactate sample, the equilibrium concentrations of HLac and H₂O are 0.32 M and 1.2 M respectively. If the previously estimated HLac:2 HDEHP stoichiometry^{††} obtains here, the extractant phase composition would be approximately 0.32 M HLac(HDEHP)₂, 0.18 M (HDEHP)₂, 1.2 M H₂O in 1,4-DIPB. Complex movements of solute molecules (including H₂O in the organic phase) appear to be an essential feature of conventional TALSPEAK chemistry.

FIGURE 11 NEAR HERE

Overall, these results clearly establish the following correlations:

1. Lactate and lanthanide partitioning are not connected under all conditions
2. Sodium lactate does not appear to be a major contributor to lactate partitioning to the organic phase, though there is some partitioning of Na⁺
3. There is a considerable salting out effect associated with the presence of higher concentrations of lactate in the aqueous phase as evidenced by increased HLac presence in the extractant phase as pH is increased (at higher pH, the [HLac]/[Lac⁻] decreases, i.e., the relative concentration of protonated lactate decreases while the salting agent NaLac increases

^{††} Evidence is offered in the discussion of neutron scattering results below that this stoichiometry might rise to HLac:HDEHP₃ under some conditions,

4. Under none of the studied conditions does the organic phase give the appearance of being saturated with lactate
5. Lactate partitioning increases as [HDEHP] increases (this could be attributable to the increased polarity of the extractant phase, but Figure 10 suggests a possible HLac:HDEHP stoichiometry of 1:2)

It also appears likely that lactic acid extraction could account for a large component of the observed pH dependence of lanthanide extraction in conventional TALSPEAK extraction systems.

Na⁺ Partitioning

The results of the lactic acid partitioning study suggested the possibility of partitioning of Na⁺ independently of lactate. In fact, lactate partitioning seems to indicate that NaLac is not partitioned in conventional TALSPEAK. Robinson and co-workers⁴³ on the other hand conclude that NaLac is the primary means by which lactate is partitioned into HDEHP isooctane solutions. McDowell and Coleman⁴⁴ have reported extraction equilibrium constants for extraction of Na⁺ from aqueous (nitrate/perchlorate) media into HDEHP; the predominant extracted complex is Na(DEHP)(HDEHP)₃. Again employing radiotracer techniques, the partitioning of Na⁺ between TALSPEAK aqueous and organic solutions has been investigated.⁴⁵ Distribution of Na and lactate between HDEHP diluted by DIPB and high concentration of lactate media with 2 M ionic strength was investigated. Distribution ratio of Na (D_{Na}) increased with increasing equilibrated aqueous lactate concentration (Figure 12). The mole ratio of Na to (HDEHP)₂ in the organic phase was greater than traditionally confirmed complex composition ratio, 0.5, in concentration ranges of [HDEHP] > 0.5 M, [lactate] > 0.7 M and pH > 3.6. These experimental results indicated the possibility for the formation of Na-lactate-HDEHP complexes in the organic phase. However, extraction model including the ternary complexation did not fit the Na and the lactate experimental results simultaneously with any combinations of stoichiometric coefficients for the ternary components.

In the figure below and absent any additional lactate, comparatively weak extraction of Na⁺ by HDEHP is indicated; the limiting slope (at low pH) of the log D vs. pH curve indicates the expected increase in Na⁺ extraction with decreasing [H⁺], but at an apparent reduced dependence

on $[H^+]$. Introduction of 1 M total lactate into the 2.0 M nitrate system has no effect on Na^+ partitioning at pH 1.3, though extraction increases with an approximate 1:1 ratio between D_{Na} and pH up to a pH of about 3.2. Above this pH D_{Na} continues to increase, but with a reduced stoichiometry with respect to $[H^+]$. A wide range of additional tests provided limited guidance relative to the establishment of an overall mechanism for Na^+ extraction. An indication of a lactate-induced specific ion activity effect can be inferred from these results, as discussed further below.

FIGURE 12 NEAR HERE

Specific Ion Activity Effects

The Na^+ extraction results and the results of other TALSPEAK thermodynamics studies suggested a contribution from cation-specific activity effects, as shown in Figure 13³². It can be seen that lactate partitioning into 0.02 M HDEHP/1,4-DIPB varies with the identity of the cation in the metal nitrate supporting electrolyte (at pH 3.6). At lactate concentrations below 0.5 M total lactate (note that at pH 3.6 lactate is approximately 50% MLac/50%HLac) Na^+ and NH_4^+ tend to suppress lactate partitioning while Li^+ and K^+ allow about ten times higher extraction of lactate. At concentrations above 0.5 M, Na^+ , K^+ , and NH_4^+ appear to promote extraction of lactate, while Li^+ does not seriously impact lactate partitioning. At 2 M lactate the order of lactate partitioning from MLac media is $NH_4^+ = Li^+ < Na^+ < K^+$. The respective hydration enthalpies are -132.5 kJ/mol, -278.5 kJ/mol, -240 kJ/mol, -252.4 kJ/mol; thus there appears to be no simple correlation between the extractability of the cation (which might be expected to correlate with the ease of dehydration of the cation) and the observed lactate partitioning. Figure 9 above has also demonstrated that Na^+ extraction does not correlate with the concentration of lactate present at pH 3.6. It could be inferred from these results (and those above) that MLac is an effective salting agent promoting the extraction of lactic acid into HDEHP/1,4-DIPB media.

FIGURE 13 NEAR HERE

To explore the effect of the medium on lactate aqueous chemistry, Zalupski and co-workers^{46,47} determined lactate stability constants for Nd^{3+} and Am^{3+} and investigated the thermochemistry of lactate protonation reactions, measuring in particular the heat of protonation of lactate in a 1 M Na lactate solution. By comparing the thermodynamic features of lactate protonation as a

function of the supporting electrolyte (using titration microcalorimetry) they concluded that the closest mimic of lactate ion in a concentrated salt medium was triflate ($\text{F}_3\text{C-SO}_3^-$). Protonation heats measured in a titration with 0.1 M HClO_4 /0.9 M NaClO_4 of 1 M NaLac and 0.01 M NaLac/0.99 M NaClO_4 resulted in ΔH_h values that differed by a factor of three. In parallel experiments with nitrate, triflate and methylsulfonate as supporting electrolyte, it was demonstrated that the thermochemical features of triflate most closely mimic those of lactate. It was also demonstrated that the background electrolyte has minimal influence on the thermodynamics of Nd^{3+} and Am^{3+} lactate complexes. Given the apparent “thermal equivalence” of Lac^- and $\text{F}_3\text{C-SO}_3^-$, it would seem reasonable to suggest that triflate could be considered a preferred non-complexing representative of lactate in investigations of lactate-based TALSPEAK solution chemistry.

Activity coefficients are usually assumed to be constant in solvent extraction systems and the typical success of application of slope analysis methods generally justifies this assumption, at least as a first approximation. In fact, in many solvent extraction systems considerable uncertainty arises when activity corrections are called for, in particular because solvent extraction reactions include both aqueous and organic media (and the intent of mixing solute species between phases) knowing where and when to apply activity corrections can be challenging. The salting out effects described above clearly indicate an opportunity for introducing activity corrections in TALSPEAK aqueous media. However, the complexity of the aqueous solution in TALSPEAK leaves the question of applying such corrections somewhat problematic. In principal, the TALSPEAK organic phase is far less complex, consisting of a two component mixture HDEHP dimers (the predominant form in aliphatic hydrocarbon solvents) and the diluent. Of course the analysis of lactate partitioning discussed above implies that even this simplification is also arguably an oversimplification. Nonetheless, Nilsson and coworkers⁴⁸ have reported activity coefficients for HDEHP in toluene, dodecane and cyclooctane using vapor pressure osmometry. They cite the utility of vapor pressure osmometry based on calibrations using multiple standards.

Speciation of Extracted Complexes

By convention in solvent extraction, slope analysis methods are considered the first option for defining the stoichiometry of extracted metal complexes in solvent extraction. As a first step, slope analysis experiments when properly constrained and performed (conducted with

appropriately preconditioned organic and aqueous phases at controlled, constant temperature and with the metal ion present at low concentrations relative to the extractant with extraction conducted from an aqueous medium in which activities of ions can be considered constant) can provide useful guidance for defining the stoichiometry of the metal complex. Several reports of TALSPEAK systems conducted using radiotracer methods and low metal loading appear to indicate a reduced extractant dependence near 2,¹⁹ which has been interpreted to indicate the extraction of $\text{Ln}(\text{Lac})(\text{DEHP}\cdot\text{HDEHP})_2$ as the dominant extracted species. Based on the observation above and the probable competition for HDEHP by both Na^+ and HLac, the first principle of a poised and understood biphasic system is probably not realistic in studies of TALSPEAK.

Recognizing this potential limitation in interpreting TALSPEAK speciation, Grimes and co-workers⁴⁹ examined the speciation of extracted lanthanide metal ions in TALSPEAK organic phases using FTIR spectroscopy and time resolved fluorescence spectroscopy. FTIR can be used to probe the organization of molecules in the extractant phase while fluorescence spectroscopy directly probes the primary coordination sphere of selected lanthanide and actinide ions. In this study, the properties of TALSPEAK organic phases created by extraction experiments of HDEHP and of (2-ethylhexyl)phosphonic acid mono(2-ethylhexyl)ester (HEH[EHP]) in 1,4-DIPB. HEH[EHP] is a more basic structural analog to HDEHP. It will emerge further along in this discussion as a potential alternative extractant to HDEHP in the Advanced TALSPEAK process. The objective of this study was to provide fresh insights into the specific role of lactate in the coordination chemistry of lanthanides in the TALSPEAK organic phase. In particular, the objective was to establish conditions under which the mixed $\text{Ln}(\text{Lac})(\text{DEHP}\cdot\text{HDEHP})_2$ mixed complex might be present. The water content in the organic phase was also monitored.

Emission spectra of Eu^{3+} complexes with HDEHP or HEH[EHP] demonstrate that, while $[\text{Eu}^{3+}]$ remains well-below the level of saturation of the organic extractant solution (1:6 Eu:HDEHP), the Eu^{3+} coordination environment remains constant as both $[\text{HLac}]_{\text{org}}$ and $[\text{H}_2\text{O}]_{\text{org}}$ increase. The fluorescence decay rate also correlates with the presence of water molecules in the coordination sphere of the Eu^{3+} metal center; as water molecules enter the cation coordination sphere the lifetime of the fluorescence decay gets shorter as the number of coordinated water molecules increases. The observed lifetimes in the extracted $\text{Eu}(\text{DEHP}\cdot\text{HDEHP})_3$ species are consistent with

the interpretation that there are no coordinated water molecules in the extracted complex under these conditions, as evidenced by the comparatively low decay constants in Figure 14. As the concentration of lactate in the organic phase increases (samples were prepared by contact of the extractant phase with aqueous media containing 0-2 M total lactate at pH 3.6), the spectral signature remains constant. Since the baseline sample was prepared in the absence of lactate, it was concluded that the most probable stoichiometry of the extracted complex was $\text{Eu}(\text{DEHP} \cdot \text{HDEHP})_3$ and that no lactate ions were coordinated to the Eu^{3+} metal center in the extracted complex, contradicting the previous reports from slope analysis results.

FIGURE 14 NEAR HERE

When higher concentrations of optically transparent La^{3+} are added to the aqueous feed to raise the total lanthanide concentration toward the saturation limit in the extracted complex while keeping $[\text{Eu}^{3+}]$ constant, the emission spectra get more complex (Figure 15)^{††}. The $^5\text{D}_0 \rightarrow ^7\text{F}_1$ transition singlet splits into a doublet with a notable increase in the intensity of both $^5\text{D}_0 \rightarrow ^7\text{F}_1$ and $^5\text{D}_0 \rightarrow ^7\text{F}_2$ electronic transitions. The increased multiplicity in the emission spectra indicates that Eu^{3+} ions may be present in multiple coordination environments at high metal loading. The increased emission intensity of the 614 nm band can be interpreted to indicate an overall decrease in the symmetry of the extracted Eu^{3+} complex in the presence of macroscopic La^{3+} . Although $[\text{H}_2\text{O}]_{\text{org}}$ increases to above 1 M at high $[\text{HL}]_{\text{tot}}$, this water does not under any conditions appear to be directly coordinated to the Eu^{3+} metal center. It appears likely that the organic phase water is associated with either HDEHP dimers, with some extracted lactic acid species, or simply solvated within interstices in the extractant solutions; the water/lactic acid extraction results described above also indicated notably high organic phase water concentrations, hence they are consistent with this suggestion. Given the high concentration of organic phase water, it is not unreasonable to project that in the presence of lactate a reorganization of HDEHP from the conventional dimer to some higher order state of aggregation (reverse micelle) might be occurring; the fluorescence spectroscopy does not directly or indirectly probe such interactions, only the state of the metal center.

^{††} It is assumed in these experiments that La^{3+} and Eu^{3+} occupy similar coordination sites in whatever complexes are present.

FIGURE 15 NEAR HERE

IR Spectra

After confirming spectrometer performance by benchmarking spectral properties in comparison with earlier work on HDEHP from Peppard and Ferraro,⁵⁰ TALSPEAK-relevant samples were prepared for examination using infrared spectroscopy by Grimes et al.⁴⁹ For the organic phase samples extracted from lactic acid buffer, when La^{3+} metal concentrations were at 0.003 M and 0.010 M, the 1680 cm^{-1} band shifted to 1740 cm^{-1} , is much sharper and displays greater intensity than the broad 1680 cm^{-1} band that is observed for the neat HDEHP sample (Figure 16). As the metal concentration is increased to 0.017 M and 0.025 M, the 1740 cm^{-1} band decreases in intensity and a new band is observed at around 1570 cm^{-1} . The new band grows in intensity as a function of metal concentration. The appearance of the two new peaks creates an isosbestic point indicating the presence of two species in the La^{3+} metal-loaded organic phases. Simultaneously with the appearance of the 1570 cm^{-1} band, the intensity in the P-O stretch (around 1200 cm^{-1}) and asymmetric stretch bands (around 1010 cm^{-1}) decreases. These combined results were interpreted to indicate the probable exchange of one or more HDEHP molecules for a lactate ion in the extracted metal complex. The IR spectra in Figure 16, panel a, (La^{3+} loaded into HDEHP from an aqueous phase containing nitric acid) does not show any change to the major absorption bands over the entire range of extracted La^{3+} metal to the organic phase. The 1680 cm^{-1} band was slightly shifted to 1650 cm^{-1} ; the P-O stretch and asymmetric stretch bands are unaffected by the increased organic phase metal concentrations.

The IR spectra indicate the lactate ion is responsible for the changing 1680 cm^{-1} band and the reduction of the P-O stretch and the asymmetric stretch bands. The isosbestic point created by increasing metal concentration in the organic phase could represent the shift of the six-coordinate $\text{La}(\text{DEHP}\cdot\text{HDEHP})_3$ complex to a ternary complex that could be $\text{La}(\text{Lac})(\text{DEHP}\cdot\text{HDEHP})_2$ (consistent with an NMR/ESI-MS study that is discussed below). Kosyakov and Yerin¹⁹ have previously postulated the presence of the mixed extracted complex $\text{Ln}(\text{Lac})(\text{DEHP}\cdot\text{HDEHP})_2$ for both Eu^{3+} and Cm^{3+} . The IR data supports the premise that as the ratio of the molar concentration of HDEHP to that of La^{3+} drops below the 6:1 ratio required for the formation of the $\text{La}(\text{DEHP}\cdot\text{HDEHP})_{3\text{ org}}$ complex, it is likely that lactate ion is integrated into the extracted complex. From the perspective of the thermodynamics of the phase transfer reaction, it matters

not whether $\text{La}(\text{Lac})^{2+}$ is formed in the aqueous phase then extracted by two HDEHP dimers or if some species of $\text{HLac}_{(\text{org})}$ complexes $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2^+$ in the interface with the coordinated release of H^+ to the aqueous phase.

FIGURE 16 NEAR HERE

NMR/ESI-MS Investigation of Speciation in TALSPEAK Extractant Phase

As a complement to the optical spectroscopy, which primarily probes the metal coordination environment, application of ^{31}P NMR spectrometry can be used to interrogate the organization of extracted lanthanide complexes from the perspective of the ligand and its arrangement around the metal center. In sufficiently robust systems, electrospray ionization-mass spectrometry (ESI-MS) can provide higher order structural information, up to and including potentially the identity of multiple species in a mixture. The adequacy of these techniques for the task objective and the value of the information and insights gained are often linked closely to chemical inertia of the bonds that hold the chemical system together. Because there is so little covalency in the bonding of lanthanide coordination complexes relevant to solvent extraction separations the issue of bond lability is an important consideration in the ultimate interpretation of the results obtained.

The optical spectroscopy results have established that under a moderately wide range of conditions the predominant extracted complex in TALSPEAK extractant phases is $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$, in which the metal center is surrounded by three extractant dimers in a pseudo-octahedral local geometry. It is important to remember the predominantly electrostatic bonding mode of these (and most other) lanthanide coordination complexes. There is evidence that the coordination environment around the metal center changes as higher concentrations of lanthanide ions are introduced. The FTIR spectra indicate that lactate is likely incorporated in these species. Investigation of some of the details of metal-extractant interactions would help establish details about the connectivity of these complexes.

Marie et al.⁵¹ completed a detailed investigation of TALSPEAK-relevant lanthanide-HDEHP interactions using ^{31}P NMR spectrometry and ESI-MS. The application of these techniques to solutions similar to those addressed in the optical spectroscopic investigation produced insights into the numbers and arrangements of HDEHP extractant molecules around the lanthanide metal

center^{§§} in the extracted complexes formed in connection with lanthanide extraction in TALSPEAK. The ³¹P NMR experiments directly probe the number of unique phosphorus environments present in the sample. The carefully-conducted ESI-MS studies provide clear indications of the metal complex species likely dominant in TALSPEAK organic phases under conditions of high metal loading. The NMR experiments also provide insights into some aspects of extractant phase ligand-exchange dynamics, which will be discussed below.

In Figure 17 five successive spectra are shown. Spectrum (1) demonstrates the narrow single resonance attributable to the HDEHP dimer, which is the dominant species present in the neat extractant solution (in 1,3-DIPB) exhibiting a chemical shift of approximately -1 ppm (relative to an H₃PO₄ external standard). Extraction of La³⁺ from HNO₃ aqueous media results in a broad composite resonance (-4 ppm) attributed to an average peak representing rapid exchange between (HDEHP)₂ and La(DEHP·HDEHP)₃; the dynamics of the exchange will be discussed below. Spectrum (3) profiles the La-HDEHP partitioning in a sample prepared under TALSPEAK – representative conditions. In this sample there is no evidence for free (HDEHP)₂ or La(DEHP·HDEHP)₃; the spectrum is characterized by two (or possibly three) closely spaced ³¹P resonances centered at about -10 ppm. The width of the lines indicates that the complexes are substitution-inert on the NMR timescale. Extraction of Sm³⁺ from solutions (Spectrum (4)) shows two relatively broad resonances representing (HDEHP)₂ and Sm(DEHP·HDEHP)₃, indicating somewhat slower ligand exchange dynamics for this system than is seen in the La³⁺ system (and some paramagnetic line broadening). The Sm-TALSPEAK sample (Spectrum (5)) shows some evidence for free (HDEHP)₂ and Sm(DEHP·HDEHP)₃ plus a third cluster of three ³¹P peaks intermediate between the (HDEHP)₂ and Sm(DEHP·HDEHP)₃ resonances.

FIGURE 17 NEAR HERE

To examine the relationship between La³⁺ extraction, aqueous lactate concentration, and the resulting distribution of HDEHP species in the extractant phase, Marie also investigated systems prepared using aqueous media containing increasing concentrations of lactate (Figure 18). Each sample was prepared by contact with a 0.025 M solution of La³⁺; at zero added lactate, the

^{§§} Due to the influence of paramagnetic lanthanide ions on the chemical shifts of NMR-active nuclei, most of the reported work focused on La³⁺, Sm³⁺ (a 4f⁵ ion that exhibits low magnetic susceptibility, hence minimal paramagnetic shifts), Y³⁺ (a diamagnetic non-lanthanide most similar in size to Ho³⁺), and Lu³⁺.

extractant phase contains only $(\text{HDEHP})_2$ and $\text{La}(\text{DEHP}\cdot\text{HDEHP})_3$ and about 0.004 M La^{3+} (Spectrum (1)). Contact with 0.5 M lactate solution increased La^{3+} extraction to near quantitative (0.024 M), accompanied by a diminished amount of $(\text{HDEHP})_2$ and $\text{La}(\text{DEHP}\cdot\text{HDEHP})_3$ and more than 90% of HDEHP in the substitution-inert peaks near -10 ppm. As the equilibrated lactate concentration increases from 0.5 M to 2.0 M, the amount of extracted La^{3+} decreases slightly (probably an indication of increased hold back of La^{3+} in the aqueous phase as higher order water soluble La-Lac complexes) and the integrations of peaks at 9.5 and 10.3 ppm apparently shifting more towards the 10.3 ppm peak. Because less La^{3+} is extracted, the free-extractant/composite peak also increases in intensity. The changing ratio of 9.5 and 10.3 ppm peaks with the change in $[\text{Lac}]$ implies direct involvement of Lac^- in the extracted complex when the extracted metal concentration is greater than the nominal 1:6 stoichiometric limit of $\text{La}(\text{DEHP}\cdot\text{HDEHP})_3$, but the integrations are not accurate enough to provide believable quantitative information.

FIGURE 18 NEAR HERE

Similar conclusions can be drawn from Figure 19, in which the lactate concentration is fixed at 1.0 M and $[\text{La}]_{\text{org}}$ varies from 3.3 to 42.5 mM. Using radiotracer (^{14}C -labeled lactate) methods, Marie and co-workers were also able to establish that in the saturation limit the $\text{La}^{3+}:\text{Lac}^-$ stoichiometry was 1:1. To satisfy the requirement of electroneutrality in the extracted complexes, it is reasonable to assume that two singly ionized HDEHP dimer-equivalents are incorporated in the mixed ligand complex.

FIGURE 19 NEAR HERE

ESI-MS Characterization of La-HDEHP Extracted Species. The possible formation of substitution-inert La^{3+} and Sm^{3+} complexes (involving $\text{La}^{3+}/\text{Sm}^{3+}$, Lac, and HDEHP in an apparent 1:1:2 mole ratio configuration) suggested a possible opportunity for effective application of ESI-MS in characterization of the species present. Marie et al. also report the results of an ESI-MS study of this system. The samples investigated were identical to those described in Figure 19 (with the radiotracer lactate omitted). Electrospray ionization (ESI) is a soft ionization technique that allows the sample molecules to be transferred directly from the solution to the gas phase by generating charged ions without appreciably disrupting metal-ligand

bonds (in the best of circumstances). For analysis, samples of the extractant solutions were diluted in methanol to facilitate volatilization. In samples of the extractant alone, monomer HDEHP was identified as the dominant species present in the analyte; the polar methanol is responsible for the de-dimerization. For samples loaded with La from nitric acid media, the predominant peaks in the mass spectrum were $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2^+$, $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2(\text{HDEHP})^+$, $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2(\text{HDEHP})_2^+$ and $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2(\text{HDEHP})_3^+$. The $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2(\text{HDEHP})_2^+$ ion (corresponding to the protonation of the 1:6 complex $\text{La}(\text{DEHP}\cdot\text{HDEHP})_3$) had the highest intensity, hence might be considered the dominant species in the bulk solvent. That these peaks persist in the methanol-diluted sample indicates a greater degree of bonding inertia in the gas phase for the lanthanide complexes than is seen for $(\text{HDEHP})_2$.

Numerous additional peaks were identified in the sample loaded from a 1 M lactate buffer, including La-DEHP binary species ($\text{La}(\text{HDEHP})\text{DEHP}_2^+$, $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2^+$, $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2(\text{HDEHP})^+$, $\text{La}(\text{DEHP}\cdot\text{HDEHP})_2(\text{HDEHP})_2^+$, $\text{La}_2(\text{HDEHP})\text{DEHP}_5^+$) and ternary species ($\text{La}(\text{DEHP}\cdot\text{HDEHP})\text{Lac}^+$, $\text{La}_2\text{DEHP}_4\text{Lac}_2\text{H}^+$, $\text{La}_2\text{DEHP}_4\text{Lac}^+$, $\text{La}_2\text{DEHP}_5\text{LacH}^+$, $\text{La}_2(\text{HDEHP})\text{DEHP}_5\text{LacH}^+$). The ternary complex ions were subjected to MS-MS analysis to develop additional information on connectivity. A collision energy of 60 V (or less) was adequate to remove ethylhexyl groups from the HDEHP core. For the most abundant species ($\text{La}_2\text{DEHP}_4\text{Lac}^+$) successive removal of eight ethylhexyl groups and a lactate left a residual peak at $m/z = 664.9$, which can be attributed to the cation $\text{La}_2(\text{PO}_4)_4\text{H}_7^+$. This species demonstrates the significant stability of the bimetallic structure formed between the two cations and the four phosphoric acid groups (PO_4^{3-}) of DEHP^- . It was suggested that the multiple ^{31}P resonances seen in the NMR spectrum at high metal loading could represent some combination of chelating, monodentate, bidentate, and bridging phosphate groups in HDEHP complexes, as postulated in Figure 20. Significantly more solution phase data would be required to develop equilibrium models (assuming that reversible equilibria are applicable in such systems) of this complex system. Results derived from both techniques used in this study are consistent with the possible formation of polymer aggregates wherein the HDEHP could bind to the lanthanide ion in a bidentate fashion or form bridges between two metal centers. Such species are consistent with the observation of different phosphorus environments by ^{31}P NMR spectroscopy.

Figure 20 NEAR HERE

Supramolecular Organization in TALSPEAK Extractant Phases Using SANS

Neutron scattering techniques have been employed to define the supramolecular organization of extractant molecules and extracted complexes in solvent extraction systems. Using the unique properties of neutrons and introducing a deuterated contrast medium (e.g., xylene-d₁₀) provide an opportunity to reveal details about the organization of solute molecules in extractant solutions. Grimes and co-workers⁵² investigated the organization of HDEHP-based TALSPEAK systems using the methods and mathematics of Small Angle Neutron Scattering (SANS) focusing on some of the same systems investigated by Marie and co-workers. The neutron scattering experiments were conducted at the High Flux Isotope Reactor at Oak Ridge National Laboratory.

As has been noted in the narrative above, lanthanide extracted complexes in TALSPEAK organic media appear to include the conventional 1:6 complex (Ln(DEHP·HDEHP)₃) at low concentrations of total metal ion, and at least one polynuclear complex with the apparent 2:2:4 (Ln:Lac:DEHP) stoichiometry as the lanthanide:HDEHP ratio drops below 1:6 (when lactate is available). The dominant form of free HDEHP in the presence of extracted lactic acid is the conventional dimer (HDEHP)₂, but there is enough HLac extracted to have an impact on [HDEHP]_{free}; some of the information on HLac extraction above suggests a possible terminal stoichiometry for extracted lactic acid of HLac·(HDEHP)₂₋₃ as an important species. Each of these species could be expected to contribute to the neutron scattering profile of TALSPEAK extractant phases.

Grimes et al. reported the results of neutron scattering experiments on extractant solutions (primarily HDEHP, but one sample contained HEH[EHP]) in deuterated xylene samples prepared for analysis by contact with buffered lactate solutions of increasing concentrations (Table 7). Parallel samples examined the influence of increasing concentrations of lanthanide ions (La, Gd, Yb taken to represent light, middle, and heavy members of the series) on the state of aggregation in TALSPEAK organic solutions. In the extractant samples equilibrated only with lactic acid, the ratio of HDEHP:HLac ranges from 200:1 down to 2.4:1 at the highest extracted lactate system. In the metal loaded samples, the ratio of metal:lactate averages near to 1 (except at the lowest lactate

concentration) while the HDEHP:Ln ratios vary from a maximum near 30:1 down to 3:1 in sample La_{0.031}. The spectrophotometry, NMR and ESI-MS results point towards domination of metal speciation by Ln(DEHP·HDEHP)₃ at low metal loading and most lactate concentrations, shifting toward domination by Ln₂(Lac)₂(DEHP)₄ as the ratio of HDEHP:Ln drops below 6:1. Comparing samples HL_{1.0} (HDEHP) with HL_{1.0} (HEH[EHP]), the 50% greater extraction of lactate and 30 times higher water content of the HDEHP samples are notable.

TABLE 7 NEAR HERE

The neutron scattering results were first analyzed considering the scattering particles to be spherical (Guinier analysis, TABLE 8^{***}). As the neutron scattering results reflect most clearly the contrast between areas of concentration of ¹H atoms and ²H atoms, the dimensions of metal ions, oxygen, phosphorus and carbon are unimportant aside from the effect they each have on ordering the orientation of the hydrogen atoms of HDEHP (primarily) relative to the deuterium of the xylene diluents, i.e., they do not impact neutron scattering directly. The fits of the scattering curves produce an estimate of the average radius of the scattering sphere (of ¹H atoms), which is used to calculate a number density of scattering particles (M_w). From the analysis the simple model can be used to calculate the number of HDEHP monomers present in the average scattering particle (n_w).

TABLE 8 NEAR HERE

It is seen in Table 8 that (HDEHP)₂ is likely the primary scattering particle⁺⁺⁺ for samples HA_{0.1}, HA_{0.2} (samples not contacted with lactate buffer solutions) and HL_{0.1, 0.5, 1.0, 1.0, HB}. A 30% larger average scattering particle is indicated for HL_{2.0}, coinciding with a 1:5 ratio of [Lac]:[HDEHP] in the extractant solution. Assuming a simple two component model (ignoring the possible contribution of monomer HDEHP to the scattering pattern), a mass balance expression can be used to relate the size of the average scattering particle to these species:

^{***} The reader is referred to the original report for the details of the mathematical treatment of SANS data by this approach.

⁺⁺⁺ Maximum scattering is provided by HDEHP based on the H-atom count in the principal scattering entities, H₂O, H₅C₃O₃, H₃₅C₁₆PO₄

$$n_w^{\text{calc}} = \frac{m^2 \cdot [\text{HLac}(\text{HDEHP})_m] + 2^2 \cdot [(\text{HDEHP})_2]_{\text{free}}}{[\text{HDEHP}]_{\text{tot}}} \quad (8)$$

where $[(\text{HDEHP})_2]_{\text{free}}$ represents all HDEHP not associated with lactic acid and m is the average HLac:HDEHP stoichiometry. Considering values of $m = 1-3$, the best fit was provided using $m = 3$ ($n_w^{\text{calc}}(\text{HL}_{0.1}) = 2.01$, $n_w^{\text{calc}}(\text{HL}_{0.5}) = 2.05$, $n_w^{\text{calc}}(\text{HL}_{1.0}) = 2.14$, $n_w^{\text{calc}}(\text{HL}_{2.0}) = 2.62$). At the highest $[\text{HLac}]_{\text{org}}$, the 0.2 M HDEHP solution is 0.041 M $(\text{HLac}) \cdot (\text{HDEHP})_3 / 0.038$ M $(\text{HDEHP})_2$ or roughly 60% bound to lactic acid, 40% free dimer.

In the metal containing samples, larger particles are seen and the complexes $\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_3^{\ddagger\ddagger}$ and $\text{Ln}_2(\text{L})_2(\text{DEHP})_4$ must be considered (besides the HDEHP and Lac species. Three mass balance expressions must be accommodated to analyze the speciation:

$$[\text{Ln}]_{\text{t org}} = [\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_3] + 2 \cdot [\text{Ln}_2(\text{L})_2(\text{DEHP})_4] \quad (9)$$

$$[\text{HLac}]_{\text{t org}} = [\text{HLac}(\text{HDEHP})_3] + 2 \cdot [\text{Ln}_2(\text{Lac})_2(\text{DEHP})_4] \quad (10)$$

$$[\text{HDEHP}]_{\text{t org}} = 2 \cdot [(\text{HDEHP})_2] + 3 \cdot [\text{HLac}(\text{HDEHP})_3] + 6 \cdot [\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_3] + 4 \cdot [\text{Ln}_2(\text{Lac})_2(\text{DEHP})_4] \quad (11)$$

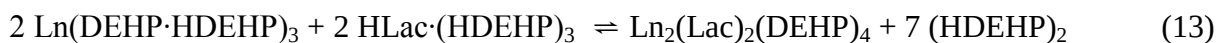
As above, HDEHP represents the primary scatterer of neutrons in these samples. Taking these relationships and species into account, the modified expression for a calculated n_w^{calc} when lanthanides are present would be:

$$n_w^{\text{calc}} = \frac{6^2 \cdot [\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_3] + 2^2 \cdot [(\text{HDEHP})_2]_{\text{free}} + 3^2 \cdot [\text{HL}(\text{HDEHP})_3] + 4^2 \cdot [\text{Ln}_2\text{L}_2(\text{DEHP})_4]}{[\text{HDEHP}]_{\text{tot}}} \quad (12)$$

In sample $\text{La}_{0.003}$, Guinier analysis indicates $n_w = 2.66 \pm 0.32$ for a sample containing 0.003 M La^{3+} , 0.0048 M HLac, 0.1004 M HDEHP. The fluorescence and NMR results suggest that $\text{La}(\text{DEHP} \cdot \text{HDEHP})_3$ should be the dominant metal complex in this lanthanide concentration range; iterative adjustment of the fit of the data to equation 12 (constrained by equations 9-11)

^{†††} Lumetta and coworkers⁵³ have recently reported results from vapor pressure osmometry and radiometric slope analysis studies that have been interpreted to indicate that the predominant lanthanide complex in HDEHP solutions in n-dodecane is $\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_2(\text{DEHP})$. The primary difference between that work and the present is the diluent, which likely accounts for this apparent difference in complex stoichiometry. In the interpretation of the present results, earlier NMR, ESI-MS, and lanthanide fluorescence results provide convincing evidence for the correctness of the $\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_3$ stoichiometry in the aromatic diluent used in the present work.

results in $n_w = 2.65$ with partitioning of La between $\text{La}(\text{DEHP} \cdot \text{HDEHP})_3$ (70%) and $\text{La}(\text{Lac})_2(\text{DEHP})_4$ (30%). In samples $\text{Yb}_{0.005}$ ($n_w = 2.70 \pm 0.33$) and $\text{Gd}_{0.013}$ ($n_w = 3.50 \pm 0.42$), the metal ion partitioning is analogously 45% $\text{Yb}(\text{DEHP} \cdot \text{HDEHP})_3$ and $\text{Yb}(\text{Lac})_2(\text{DEHP})_4$ (30%) and $\text{Gd}(\text{DEHP} \cdot \text{HDEHP})_3$ (33%) and $\text{Gd}(\text{Lac})_2(\text{DEHP})_4$ (67%). Based on these analyses, 75% of the scattering particles are $(\text{HDEHP})_2$ in $\text{La}_{0.003}$ and $\text{Yb}_{0.005}$, 45% in $\text{Gd}_{0.013}$. With abundant lactate available, $\text{Ln}(\text{Lac})_2(\text{DEHP})_4$ becomes increasingly important as the metal concentration increases. In the $\text{Gd}_{0.013}$ sample the distribution of HDEHP among the projected species is 45% $(\text{HDEHP})_2$, 26% $\text{Gd}(\text{DEHP} \cdot \text{HDEHP})_3$, 17% $\text{Gd}_2(\text{L})_2(\text{DEHP})_4$, and 12% $\text{HL}(\text{HDEHP})_3$. A balanced organic phase equilibrium constant expression can be suggested to define the relationship between these species, as follows:



however, this single expression does not produce a single equilibrium constant value for the three systems tested, suggesting that perhaps additional reactions contribute. The results of the fitting exercise are summarized in Table 9.

TABLE 9 NEAR HERE

For samples $\text{La}_{0.023}$ and $\text{La}_{0.031}$ much larger aggregation numbers are indicated thus it is reasonable to project that some degree of supramolecular ordering of these species is probable. Given the trends in the relative concentrations as the metal concentration increases, it is reasonable to assume that the dominant “monomer” species in these clusters is $\text{La}_2\text{Lac}_2(\text{DEHP})_4$ and that the larger aggregates correspond to clusters of these species, each representing about four HDEHP monomer equivalents, perhaps $[\text{La}_2\text{Lac}_2(\text{DEHP})_4]_{3-4}$.

The scattering results demonstrate that the dominant HDEHP species in the absence of metal ions is the hydrogen bonded dimer, $(\text{HDEHP})_2$. Absent lanthanides, it appears that lactic acid is extracted as the 1:3 complex $\text{HL} \cdot (\text{HDEHP})_3$. Scattering in samples containing up to 0.005 M lanthanides (prepared by extracting lanthanides from aqueous media containing 1.0 M buffered lactic acid) indicate that the dominant metal complex is $\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_3$, but $\text{Ln}(\text{Lac})_2 \cdot (\text{DEHP})_4$ is also important. At 0.013 M extracted lanthanide, the scattering results indicate lower $\text{Ln}:\text{DEHP}$ stoichiometry and larger scattering particles. At higher metal concentrations, the SANS results indicate large aggregates, the largest aggregates achieving a

size equivalent to up to 20 HDEHP monomers as the primary scattering entity^{§§§}. Analysis of particle shapes indicate best fits with a uniform oblate spheroid particle.

Phase Transfer Kinetics in TALSPEAK

The second important issue impacting the design and eventual deployment of TALSPEAK chemistry at an industrial scale is that of the reported “slow” phase transfer kinetics. It has been noted by several authors that at lactate concentrations below 0.5, the rate of phase transfer is unacceptably slow in TALSPEAK. Nilsson and coworkers⁵⁴ investigated the effect of lactate on the pattern of phase transfer rates of lanthanides and americium using a standard protocol of vortex mixing combined with either radiometric or ICP-MS detection. The results shown in Figure 21 demonstrate that for the lightest lanthanides (La, Ce, Pr) and for the weakly extracted Am^{3+} , lactate has minimal influence on the rate of phase transfer with the 30 minute mixing period. For Nd through Lu, extraction kinetics (under the conditions of these experiments) are extremely slow demonstrating poor extraction of these ions (and no separation from Am^{3+}) in the absence of lactate. At 0.5 M lactate, the extraction of Eu through Lu is adequate to provide a group separation, but is far from reaching an equilibrium partitioning level, as represented by the 1 M lactate data. The pattern of extraction seen in the 1 M lactate system closely approximates the predictions of the thermodynamic model described in the earlier review. Additional, more detailed studies of phase transfer kinetics in this system have been completed,⁵⁵ but the analysis of these studies is pending.

FIGURE 21 NEAR HERE

Because phase transfer kinetics experiments are moderately challenging to conduct, there is a comparatively limited base of experimental investigations relevant to TALSPEAK, though the book chapter by Danesi⁵⁶ provides a quality overview of the subject of solvent extraction kinetics. A recently developed approach to investigating the kinetics of phase transfer reactions in solvent extraction is the application of microfluidic devices to this task; an example of the latter will be described below.

^{§§§} There was no evidence of any tendency to form a second organic phase in any of these samples.

Of course, the challenging phase transfer kinetics in TALSPEAK separations is not a recent observation. Though there have been multiple experimental examinations of TALSPEAK phase transfer kinetics, two earlier investigations of TALSPEAK phase transfer kinetics are of particular note. Kolarik²³ examined the kinetics of extraction in conventional TALSPEAK, noting the role of lactate as a phase transfer catalyst, but ultimately ascribing the slow phase transfer kinetics to the dissociation of a $\text{Ln}(\text{DTPA})^{2-}$ complex in the interface.

Danesi and Cianetti²⁶ investigated this system using a constant interface Lewis cell and developed the following mathematical expression (and parameters) to profile the extraction of Eu^{3+} by HDEHP. These authors concluded that “...in a wide range of conditions,...mass transfer process is *entirely* controlled by the rate of *interfacial* chemical reactions.” In their mechanism, Eu^{3+} , Eu-lactate complexes, Eu aminopolycarboxylate complexes and mixed $\text{Eu}(\text{Lac})(\text{APC})$ species are considered to interact with $\text{HDEHP}_{\text{int}}$. Numerous transient mixed ($\text{Eu}(\text{Lac})-(\text{APC})-(\text{DEHP})$) complexes are proposed with (at least) ten rate constants as adjustable parameters. They do observe that HEDTA releases Ln^{3+} ions more readily than DTPA.

Starting from comparatively simple observations on HDEHP extraction, Danesi developed increasingly complex equations to describe the extraction of lanthanides into HDEHP/dodecane solutions in the absence of aqueous complexing (four rate parameters), in the presence of aqueous lactate (four independent rate parameters and three complex formation equilibrium constants), in the presence of aminopolycarboxylic acid complexes but without lactate, and finally in the presence of both lactate and an aminopolycarboxylate (17 rate constants describing forward and reverse extraction reactions). Comparing experimental results for extraction from DTPA and HEDTA containing media a ternary ($\text{Ln}(\text{Lac})(\text{APC})$) complex is postulated, with acknowledgement that the crowded environment around DTPA may well inhibit this feature of the exchange, while the ternary complex is more relevant in the less crowded HEDTA system. Finally it was noted that a two times larger mass transfer rate could be achieved at a 2.5 times lower lactate concentration in HEDTA-lactate systems than in DTPA-lactate based equivalents.

Kinetics investigations of this type are not frequently conducted due to the challenge of designing and executing meaningful experiments. Additionally, studies of TALSPEAK kinetics must at some point involve the use of radioactive materials, configured for study using conventional equipment potentially significant quantities of radioactive materials. Nichols and coworkers⁵⁷ have reported results of a test of a plug-based microfluidic system that uses flowing plugs

(droplets) in microfluidic channels to determine absolute interfacial mass transfer rate constants under conditions of both rapid mixing and controlled interfacial area. In this report, this system was used to determine the rate constants for interfacial transfer of all lanthanides, minus promethium, plus yttrium, under TALSPEAK process conditions. This development promises possible new inroads into understanding of phase transfer kinetics.

Chemical Kinetics of TALSPEAK: the Role of Lactate

It is clear from the results of these biphasic kinetics investigations that transient species existing at the phase boundary govern the rate of lanthanide cations crossing from the aqueous to organic phase in conventional TALSPEAK. It is also clear that lactate ion plays a central role in this phase transfer process, at least for TALSPEAK operated in the forward extraction mode (Reverse TALSPEAK in which Ln and An are extracted then the actinide ions selectively stripped using buffered aminopolycarboxylate complexants have received less detailed examination). Useful as they are in support of process development, these observations offer minimal opportunity for gaining insight into the molecular-scale details of the interactions. Because the phase-transfer process is inherently too complex to easily untangle details of the molecular-scale interactions, investigations have been completed of the ligand exchange dynamics of TALSPEAK-relevant organic and aqueous systems. Organic phase interactions have been approached using ^{31}P NMR spectrometry while the aqueous systems have been examined using optical methods, (stopped-flow spectrophotometry).

Extractant phase dynamics and species. Marie et al ⁵¹ report the results of an investigation of ligand exchange dynamics of HDEHP solutions containing selected lanthanide cations with and without lactate being present. All samples investigated in that study were prepared by solvent extraction, hence the organic media were saturated with equilibrium amounts of extracted H_2O (and lactic acid in the samples prepared by extraction from lactate buffer media). The methods of ^{31}P NMR spectrometry were used to investigate the rate of HDEHP exchange in the binary complexes $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$ $\text{Ln} = \text{La}$ and Sm^{****}) that represent the predominant extracted complex at low metal loading, as described above in the spectroscopic and SANS; the diluent was 1,3-diisopropylbenzene (Figure 22). The rate data indicate considerably faster ligand

**** Samarium was selected for this observation because its low magnetic susceptibility allows NMR measurements to be made without significant paramagnetic line broadening or substantial shifts.

exchange kinetics for $\text{La}(\text{AHA})_3$ than is seen for $\text{Sm}(\text{AHA})_3$, with a corresponding shift from a dissociative interchange to an associative process. The authors interpret these results to indicate that "...in the $\text{La}(\text{AHA})_3$ system ligand dimers are being exchanged, while in the more strongly interacting $\text{Sm}(\text{AHA})_3$ system monomers of HA might be exchanging. In this framework, the more weakly coordinated AHA^- dimer is partially released to a monomeric configuration with concomitant monodentate coordination of a free $(\text{HA})_2$ dimer occurring in the opposite quadrant of the $\text{La}(\text{AHA})_2^+$ core $\{\text{AHA}\cdots\text{La}(\text{AHA})_2\cdots\text{AHA}\}^\ddagger$. The comparatively large $\Delta H^\ddagger$ would arise from the energetic expense of moving larger dimers through the non-polar 1,3-dipb medium. The alternative associative mechanism in the $\text{Sm}(\text{AHA})_3$ (with small $\Delta H^\ddagger$) could be rationalized in terms of an expansion of the Sm^{3+} coordination sphere to accept a seventh monomer HDEHP $\{\text{Sm}(\text{A})_3(\text{HA})_4\}^\ddagger$ in the activated complex."

FIGURE 22 NEAR HERE

NMR spectra of paramagnetic lanthanide ion complexes indicate the expected paramagnetic shifts of bound extractant ^{31}P signals; for all lanthanide ions heavier than Sm, the $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$ complexes are substitution inert within the limits of the NMR relaxation timeframe (milliseconds) at ambient temperatures. With the introduction of lactate and at lanthanide concentrations above the stoichiometric limit of the $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$ complex, the metal complexes ultimately shift toward $(\text{Ln}(\text{Lac})(\text{DEHP})_2)_n$ stoichiometry. These complexes are all substitution inert (on the NMR scale) and typically are represented by two or three distinct phosphorus environments (as indicated by variations in the δ - values), as has been noted and discussed above.

Aqueous Complexation Kinetics. The results shown in Figure 21 clearly demonstrate the importance of lactate in the rate of the TALSPEAK phase transfer reaction. The TALSPEAK phase transfer mechanism of Danesi and Cianetti ²⁶ establishes an empirical relationship between the rate of the phase transfer reaction and lactate concentration, but it provides little guidance regarding the chemical kinetic processes. To examine more directly the origins of this phase transfer catalysis from the perspective of the DTPA complexes that are formed (and dissociated), Nash and co-workers ⁵⁸ have employed stopped-flow optical spectrophotometric methods to characterize the rate and mechanism of the reaction between lanthanide ions and DTPA (and for additional insight, EDTA) using stopped flow spectrophotometry. Following on from earlier

investigations of actinide complexation by a variety of optically transparent ligands, this application requires mixing (from separate syringes) of a dilute solution of a (strongly colored) lanthanide complex with Arsenazo III ((2,2'-(1,8-dihydroxy-3,6-disulfonaphthylene-2,7-bisazo)bis(benzenearsonic acid) with a lactate buffered solution of DTPA (or EDTA) at an appropriate pH. The lanthanide-Arsenazo III complex dissociates and the released lanthanide cation is then complexed by the stronger DTPA. The reaction is complete in the sub second(s) timeframe, as measured using an Olis RSM-1000 spectrophotometer. Determination of mechanistic details and patterns of kinetic stability across the lanthanide series are established by the systematic variation of solution properties.

Experiments were conducted as a function of EDTA/DTPA ligand concentration, total lactic acid concentration, and pH. The equilibrium perturbation reaction proceeds as a first order approach to equilibrium over a wide range of conditions, allowing determination of complex formation and dissociation rate constants in the same experiment. The rate of the complexation reaction was reported for the entire lanthanide series (except Pm^{3+}). The predominant pathway for lanthanide-EDTA and lanthanide-DTPA dissociation is inversely dependent on the total lactate concentration; the complex formation reaction demonstrates an unusual direct dependence on $[\text{H}^+]$. Unexpectedly, the rate of the complex formation reaction is seen to be fastest for Gd^{3+} (Figure 23). Correlation of these results indicates that in 0.3 M lactate solutions the exchange of lanthanide ions between lactate complexes and the polyaminopolycarboxylate govern the process.

TABLE 10 NEAR HERE

FIGURE 23 NEAR HERE

Correlations of rate data with thermodynamic equilibrium calculations lead ultimately to the conclusion that the primary reactants in the observed reaction are $\text{Ln}(\text{Lac})_n^{3-n}$ species interacting with either $\text{H}_3\text{DTPA}^{2-}$ or $\text{H}_2\text{DTPA}^{3-}$. In 0.3 M lactate buffer (pH 3.6), Gd^{3+} complex formation reactions are fastest for both ligand systems and Lu^{3+} is the slowest. This result appears to be most closely correlated with the equilibrium lactate-lanthanide complex stoichiometry, which is 2.46 at Gd^{3+} , 2.94 for Lu^{3+} . This result is taken as an indication of reaction inhibition by the formally neutral $\text{Ln}(\text{Lac})_3$ complex.

This research was continued with a detailed examination of the complexation kinetics of these same lanthanide ions with DTPA, EDTA and HEDTA in 1 M lactate buffer.⁵⁹ In this investigation, it was discovered that the reaction describing the complex formation reaction of LnDTPA^{2-} occurs under TALSPEAK conditions via the formation of an unidentified $(\text{Lac})\text{Ln}(\text{DTPA})^{x-}$ transient complex. At higher lactate concentrations the Ln-DTPA complex formation reaction slows. The activation entropy for this reaction is strongly positive, which is interpreted to require the dissociation of a lactate from a higher order $\text{Ln}(\text{Lac})_n$ species. This observation leads to the postulation of a chemical kinetics mechanism to describe Ln extraction as follows (considering only secondarily the influence of inter-phase transport reactions):



In this hypothetical reaction sequence, the first rate limiting process (reaction 1) is the release of the lanthanide cation from the thermodynamically stable $\text{Ln}(\text{DTPA})^{2-}$; this reaction is typically accompanied by rapid recombination reaction (reaction 2) and diffusion-controlled protonation of DTPA^{5-} (reaction 3). Such reactions have been reported in the prior literature as being subject to catalysis by H^+ and acetate buffer;^{24,25} the TALSPEAK-relevant work does not clearly establish H^+ involvement, though lactate ion is certainly important. Reaction 3 is not rate-process relevant since these reactions occur rapidly and spontaneously at constant pH. The high concentrations of lactate anion in TALSPEAK drives the rapid formation of 1:1, 1:2, and 1:3 complexes (as

dictated by equilibrium thermodynamics – the average ligand number varies across the series in a non-linear fashion).

The tris-lactate complex presents a moderately hydrophobic outer shell (assuming that each lactate is bidentate, the residual hydration is 2-3 water molecules) thus it can more readily diffuse toward the less polar interface double layer (reaction 7). In the interface, the de-dimerized HDEHP monomer exchanges its H^+ to a lactate ion (reaction 8) and “drags” the $Ln(Lac)_2(DEHP)$ deeper into the organic phase where the more abundant and more hydrophobic $(HDEHP)_2$ displaces the remaining lactate molecules forming the very stable and highly hydrophobic pseudo-octahedral $Ln(DEHP \cdot HDEHP)_3$ complex (reaction 9). The ESI-MS, NMR, optical spectroscopy and SANS results indicate that more complex mixed ligand complexes become more important as the metal ion concentration increases to $[Ln]:[HDEHP]$ ratios less than 1:6 (HDEHP monomer equivalents) at which point apparently polynuclear, mixed-ligand complexes become more important. In this analysis, the increased hydrophobicity of $Ln(Lac)_3$ and the lability (rapid ligand exchange) of these complexes would appear to represent a good combination for more rapid phase transfer kinetics.

Adjustments to TALSPEAK

Process Control. One approach to deployment of conventional TALSPEAK is as a component of a multiple unit operation nuclear fuel recycling program. In a modern, computer-controlled reprocessing facility, it is not unreasonable to project that the issue of pH control in TALSPEAK might be achieved by instituting a system of on-line monitoring and adjustment. Bryan and coworkers have developed a suite of on-line monitoring devices based on multiple spectroscopic techniques.⁶⁰ To translate static to flow measurements and monitor extraction processes, a centrifugal contactor has been equipped with Raman, UV/Vis/NIR spectroscopy and continuous data acquisition of temperature, flow, and density of aqueous and organic phases. This system provides immediate chemical data and flow parameters of high-level radioactive waste streams. Testing has been performed under both PUREX- and TALSPEAK-like used nuclear fuel separation processes. In addition to monitoring target analytes, methods are being developed to measure free acid/pH spectroscopically (to overcome some of the issues arising from the fragility of conventional glass electrode methods). The primary drivers for on line monitoring and control

capabilities is materials control and accountability for fissile material, but application of on-line control capability could address the pH control issues of conventional TALSPEAK.

Effect of Elevated Temperature. To address the issue of slow phase transfer kinetics in TALSPEAK, one obvious factor to consider is the possible operation of the process at elevated temperature. The Arrhenius law teaches that the rates of chemical reactions increase with temperature. Diffusion processes are also impacted by changes in temperature. Of course, in a solvent extraction reaction there are numerous processes occurring (both driving and retarding extraction), each with a distinct activation energy. Thermal effects are of interest in TALSPEAK applications in any event, as deposition of radioactive decay byproducts in principle leads to elevation of the temperature as a natural effect of the ongoing radioactive decay of components. It is therefore quite reasonable to consider the effect of elevated temperature on operation of TALSPEAK. The cautionary note regarding the effect of temperature on TALSPEAK performance is that there are so many competing reactions relevant to TALSPEAK that the net effect of a temperature increase is not easily predicted.

Tian and Rao³⁵ investigated the effect of temperature on the protonation equilibria of lactate and DTPA. The protonation constants at 10-70° were determined by titration. Potentiometry and the protonation enthalpies were determined at 25° by titration microcalorimetry. The protonation constants remain essentially unchanged (25-70°) within the experimental uncertainties, indicating that the effect of temperature on the protonation of lactate should be insignificant. In contrast, the protonation constants of DTPA generally decrease as the temperature is increased. Exothermic complexation heats discussed above suggest that it is likely that lanthanide/actinide complex stability will also decrease in parallel. These authors also have reported preliminary results on complex stability for Eu^{3+} and Cm^{3+} based on fluorescence spectroscopy.⁶¹ The smaller exothermic enthalpies associated with metal lactate complexes suggest that the competitive balance between lactate and DTPA might be altered at elevated temperature, but it is not clear how significant the impact on separation performance might be. Burns and Tokheim have conducted a series of lanthanide partitioning experiments at 50°C in lactate and citrate media, finding only slight improvements in phase transfer kinetics and under some circumstances larger uncertainties in experimental observations.⁶²

Effects of Radiation Chemistry. It is known from prior investigations of TALSPEAK that DTPA is damaged by radiolysis, but also that high concentrations of lactate exert a radiation protection effect (probably as a radical scavenger) that extends the effective life of DTPA.^{11,63} In a recent addition to the knowledge base for radiolysis effects on TALSPEAK, Martin and co-workers^{64,65} have observed that OH[•] radicals oxidized lactic acid and sodium lactate to produce a different distribution of degradation products. Gamma radiolysis of a 1 M lactate buffer does cause an increase in pH, though Eu extraction remains relatively constant. Liquid chromatographic analysis establishes the presence of numerous degradation products, including possible condensation products of high mass. The degradation products did not interfere with Eu³⁺ stripping from the loaded organic phase.

Application of TALSPEAK Chemistry in Extraction Chromatography. During the past five decades, TALSPEAK chemistry has been developed, adjusted and optimized for application in countercurrent liquid-liquid extraction applications. To gain additional insight to the role of diffusion processes and exchange kinetics on process application, Braley et al.⁶⁶ chose to examine the potential for achieving improved TALSPEAK performance by substituting extraction chromatographic materials (Eichrom LN1 and LN2 materials) for the HDEHP – 1,4-DIPB extractant. The effectiveness of alternative reagents, (HEH[EHP]) (immobilized on a support) and triethylenetetramine-N,N,N',N'',N''',N''''-hexaacetic acid (TTHA) (in the mobile phase) was also considered in this investigation. The results shown in Figure 24 indicate that the concentrated extractant in the EXC material behaves similarly to the lipophilic extractant system. It appears that, just as in the conventional TALSPEAK solvent extraction system, lactate partitioning into the resin phase consumes resin capacity. This effect reduces lanthanide/amerium extraction efficiency of the HDEHP EXC material. The weaker hold-back reagent (under TALSPEAK aqueous conditions, pH ~ 3.6), TTHA, actually provides slightly improved Am³⁺/Ln³⁺ separations performance relative to DTPA. The ultimate conclusion of this study is that TALSPEAK chemistry works as designed in this system, but the application of TALSPEAK in a chromatographic mode offers minimal advantage over solvent extraction. Lactic acid partitioning into the LN1 resin phase (based on HDEHP) reduced Ln uptake capacity; the LN2 resin (based on the more basic extractant HEH[EHP]) was less impacted by this phenomenon. This knowledge reinforced/inspired consideration of an Advanced TALSPEAK process based on this extractant, as will be discussed in the next section.

FIGURE 24 NEAR HERE

Advanced TALSPEAK/TALSQuEAK

The results of the cumulative investigations described above indicated quite clearly that lactate partitioning into the extractant phase, at a minimum increasing the complexity of TALSPEAK, was most probably the key to resolving the enigmatic pH dependence of TALSPEAK. The initial thermodynamic modeling of the conventional TALSPEAK suggested that this system should not be plagued by a steep acid dependence. Application of various characterization methods established that, while lactic acid did serve an integral role in improving phase transfer kinetics and increasing DTPA solubility and radiolytic stability, its partitioning into the extractant phase also appeared to be the best candidate for explaining the steep pH dependence of TALSPEAK through this competition for the HDEHP. Weaver and Kappelmann considered a wide array of buffers, including glycolic and citric acid, but ultimately settled on lactate. In more recent years, citrate has become the preferred TALSPEAK buffer for fast-track application development studies. One rationale for choosing citrate is its increased buffering capacity.

Expanding on the insights gained in these investigations of conventional TALSPEAK, Braley and co-workers⁶⁷ began an investigation of the possible advantages to be gained in TALSPEAK-based separations by shifting to a “weaker” (more basic) extractant molecule balanced against a weaker actinide-selective holdback reagent. It was reasoned that the increased basicity of a phosphonic acid extractant HEH[EHP] would decrease the tendency of lactic acid (or other buffer) to partition into the extractant phase, eliminating HLac competition for the extractant and quite possibly relieving the steep acid dependence of conventional TALSPEAK systems based on HDEHP. Based on the knowledge gained in studying conventional TALSPEAK, it was clear that using a more basic (“weaker”) extractant like HEH[EHP] would demand a weaker actinide holdback reagent for process balance. If such a system might exhibit simultaneously reduced affinity for lactate, lactate might well remain an acceptable TALSPEAK buffering medium.

As has been noted above, the trans-lanthanide pattern of stability of DTPA is an important feature of the separation pattern seen in conventional TALSPEAK. A search of the literature revealed that both TTHA and N-(2-hydroxyethyl) ethylenediamine-N,N',N'-triacetic acid

(HEDTA) exhibited a DTPA-like pattern of increasing stability of lanthanide complexes from La to Eu followed by relative insensitivity to lanthanide cation radii for the heavy lanthanides. In the system based on a 0.2-1.0 M sodium lactate buffer matched to HEH[EHP], the undesirable partitioning of Na^+ , lactic acid, and water into the organic phase is greatly reduced relative to conventional TALSPEAK. TTHA appears to offer little advantage over DTPA in conventional TALSPEAK, but both DTPA and TTHA are too strong for use in combination with HEH[EHP]. The combination of HEDTA with HEH[EHP] achieves good balance and exhibits a nearly flat pH dependence between 2.5 and 4.5, in contrast with conventional TALSPEAK. This combination demonstrates more predictable performance than is seen in conventional TALSPEAK, while providing acceptable americium/lanthanide separation factors. The HEDTA/HEH[EHP] combination offers the additional advantage of more rapid phase transfer kinetics for the heavier lanthanides without the need for high concentrations of a lactate buffer.

FIGURE 25 NEAR HERE

FIGURE 26 NEAR HERE

Mixed Ligand Complexes in Advanced TALSPEAK. In the Advanced TALSPEAK process, the replacement of DTPA as the actinide selective holdback reagent with HEDTA will result in a reduction of the total complexant denticity from 8 to 5 coordination sites. As lanthanide ions do not exert orbital-based directional bonding interactions the reduced coordination environment of HEDTA leaves space for the addition of a secondary ligand, e.g., lactate. It is conceivable that this ability to form such ternary complexes could be employed to enhance group separations. Recognizing this potential, Braley and co-workers⁶⁸ investigated the role of carboxylic acids as supplementary ligands in TALSQuEAK separations. UV-Visible spectrophotometric studies of the Nd hypersensitive band indicates the presence of aqueous ternary Nd-Lac-HEDTA species. For the following reaction, $\text{Nd}(\text{HEDTA}) + \text{Lac}^- \rightleftharpoons \text{Nd}(\text{HEDTA})\text{Lac}^-$ an equilibrium constant of $K_{111} = 1.83 \pm 0.01$ at 1.0 M ionic strength was reported. Such complexes reduce extraction of f element ions more-or-less across the board; compensation with higher concentrations of extractant could restore extraction, again without substantial effect on inter-group separation efficiency. There is little indication that the ternary complexes based on HEDTA-lactate combinations would have a positive effect on the group separation efficiency of Advanced

TALSPEAK systems, however, adding an appropriately structures soft donor complexant could amplify group separation efficiency.

Alternative Extractants, Complexants Media and Buffers. Rudisill et al.⁶⁹ measured the distribution of lanthanide and actinide elements between TALSPEAK-like aqueous and organic phases in the presence of complexants which were potentially less sensitive to pH control than DTPA. The complexants tested included aceto-, benzo-, and salicylhydroxamic acids (a class of reagents considered integral to the Pu/Np partitioning in the UREX+ Processes), N,N,N',N'-tetrakis(2-pyridylmethyl) ethylenediamine (TPEN), and ammonium thiocyanate (NH₄SCN). The hydroxamic acids were the least effective of the complexants tested. The separation factors for TPEN and NH₄SCN were higher, particularly for the heaviest lanthanides in the series. However, no conditions were identified which resulted in separation factors that approached those measured for the use of DTPA.

Grimes and coworkers⁷⁰ have examined the potential for developing TALSPEAK systems based on amino acid buffers and operation at lower pH than conventional TALSPEAK. In this investigation the TALSPEAK aqueous phase was modified by replacing the lactic acid buffer with a variety of simple and longer-chain amino acid buffers. The results show successful trivalent lanthanide/actinide group separation with the aqueous-modified TALSPEAK process at pH 2. The amino acid buffer concentrations were reduced to 0.5 M (at pH 2), and separations were performed without any effect on phase-transfer kinetics. Successful modeling of the aqueous-modified TALSPEAK process ($p[H^+] = 1.6-3.1$) using a simplified thermodynamic model and an internally consistent set of thermodynamic data adequately models system performance. Introduction of an amino acid buffer does accomplish at least one significant improvement to TALSPEAK, as the zwitterionic amino acids resist partitioning into the extractant phase and their lanthanide complexes are notably weak, hence the chemistry is considerably simplified.

During the past 15 years, the class of reagents known as room temperature ionic liquids have been extensively investigated for many different possible applications, including their possible use in solvent extraction for metal ion separations. Schrob and co-workers⁷¹ have recently reported the results of a study in which they claim to “re-invent” TALSPEAK, retaining the

competitive ion binding but changing considerably the chemical implementation of the underlying general principles. The DTPA moiety is integrated into a functionalized ionic liq. (IL) that is immiscible with an organic phase containing dialkyl phosphate ligands. Choline and betainiumbistriflimides double as IL diluents and synthetic reagents. A deprotonated DTPA-betainium conjugate forms 1:1 complexes with trivalent f-element cations whereas these metal ions form 1:2 complexes with the DTPA-choline conjugates. Large separation factors for Eu/Am partitioning between the two phases are observed approaching 120-150 for DTPA-betainium and 250-270 for DTPA-choline. In the latter system, the traditional TALSPEAK "parabolic" dependence of the phase distribution ratios as a function of ionic radius is reported. Dai and co-workers⁷² have reported on studies investigating eight imidazolium ILs([Cnmim][NTf2] and [Cnmim][BETI], n = 4,6,8,10) and one pyrrolidinium IL ([C4mPy][NTf2]) ionic liquids as diluents using HDEHP as an extractant for the separation of lanthanide ions from aqueous solutions of 0.05 M glycolic acid or citric acid and 0.005 M DTPA. The extraction efficiencies were studied in comparison with DIPB.

Other Advanced TALSPEAK Concepts

Broadly speaking, TALSPEAK chemistry can be considered to be any solvent extraction system that is based on the application of an aqueous medium composed of a moderately concentrated carboxylic acid buffer solution that also contains an actinide-selective complexing agent. Most such applications have been developed by using a representative TALSPEAK aqueous phase in contact with most any extractant that permits efficient extraction of trivalent f-elements – as the group selectivity arises from the soft-donor holdback/stripping reagent, most any extractant phase that exhibits a suitable lanthanide extraction patterns could be successfully coupled to a buffered aminopolycarboxylate aqueous phase. In reality, the task can be simplified, as it is (from the perspective of the nuclear chemistry/physics motivations for separating the groups) only really essential to gain separation of Am (or Am and Cm) from Pr through Gd; neutron capture cross sections for La and Ce are notably less than those of Nd-Gd and fission yields are quite low for lanthanides heavier than Tb. To achieve these separations of such fundamentally similar ions remains one of the most difficult challenges in metal ion separation science. Since the problem

has not been solved and in industrial-scale production anywhere in the world, it must be considered as still an open challenge.

Another driver for advanced TALSPEAK applications was developed as a result of attempts to “collapse” the UREX concept by combining extractant operations to reduce the number of unit operations in a reprocessing scheme. In UREX, it would clearly improve the economic outlook for eventual implementation of a process if unit operations (for example TRUEX separation of f elements from d-transition elements) in used fuel and the subsequent separation of lanthanides from trivalent actinides into a single combined process. A number of such concepts have appeared in the recent literature. Process development activities around the world investigating such processes have established a variation on a lesson learned from TALSPEAK studies, that is chemical reagents that are reactive will react, often in ways that were not anticipated. In combined process research development the lesson that has been learned is that extractant molecules sometimes do the unexpected. In studies of TALSPEAK, we have learned that the unexpected extraction of lactic acid by HDEHP causes complications in process control. In the development of the TRUSPEAK process, Tkac and co-workers⁷³ discovered that the interactions between HDEHP and CMPO defeated the objective of selectively partitioning f elements. Perhaps the most important take home message from these experiences is that there is no substitute for direct examination of interactions.

Conclusions

TALSPEAK chemistry is at the same time one of the best established and most flexible methods known today for efficiently partitioning fission product lanthanides from transplutonium actinides, and an extremely complex solution chemistry challenge. The principle is simple, hold back one group while extracting a contaminant, not necessarily as efficient of an approach as selectively extracting the minor constituent (actinide) away from the major component (lanthanides), but still tractable and proven. It is in the execution of the chemistry that the process gets complicated. The complexity of solute interactions in the baseline TALSPEAK process (HDEHP in 1,4-DIPB, concentrated lactate buffer, DTPA actinide holdback reagent) leads to complications in process implementation; of particular importance are the long recognized issues of pH control and slow phase transfer kinetics. Numerous investigations of TALSPEAK and

TALSPEAK-like applications have confirmed the reality of these issues, generated possible explanations, and even provided “work-around” solutions and adjustments to cope with the effects. Until recently, comparatively few investigations of TALSPEAK have addressed the value of understanding more completely the underlying chemistry that leads to these complications.

During the past decade significant advances have been made in understanding the fundamental solution chemistry that underpins the TALSPEAK process through the application of a variety of standard analyses of conventional TALSPEAK systems. To maximize the value of the information developed, most of these experiments were designed to control those variables (ionic strength, temperature, pH) that enable the interpretation of fundamental data in terms of known principles of solution chemistry. Rather than rely on slope analysis as an analytical tool, techniques that directly interrogate the metal ions, buffer and complexing agents were employed. This approach has provided enough improved understanding of the underlying chemistry to support the possible application of TALSPEAK principles in new approaches to this difficult separation.

Concerning the principal limitations of TALSPEAK chemistry, the following new information has been developed (focusing on the baseline TALSPEAK formulation):

- Lactic acid partitioning into HDEHP/1,4-DIPB can occur independently of the extraction of lanthanides under all loading conditions, as $\text{Ln}(\text{Lac})^{2+}$ species only under conditions of high metal loading ($[\text{Ln}]:[\text{HDEHP}]$ higher than 1:6); extraction of $\text{Ln}(\text{Lac})^{2+}$ had been implicated as the primary pathway to lactate partitioning in the prior literature
- The primary extracted lactate species appears to be $\text{HLac}\cdot\text{HDEHP}_3$
- Lactate ion appears to exert a moderately strong specific ion activity (aqueous phase salting-out) effect, significantly impeding thermodynamic modeling efforts
- Na^+ is extracted, but it is not clear whether this is direct extraction of Na^+ by $(\text{HDEHP})_2$ or extraction of NaLac , or both; lactate ion activity effects complicate interpretations of

Na^+ extraction; Li^+ , Na^+ , K^+ , and NH_4^+ exhibit different extraction efficiencies, but Na^+ , K^+ , and NH_4^+ each show evidence of lactate salting out effects

- $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$ appears to be the primary extracted complex in TALSPEAK at $[\text{Ln}]/[\text{HDEHP}]$ ratios below the stoichiometric limit of 1:6; as the metal concentration increases, an apparent HDEHP-bridged dinuclear complex ($\text{Ln}_2(\text{Lac})_2(\text{DEHP})_4$) becomes increasingly important – at high lanthanide concentrations supramolecular organization of solutes occurs producing aggregates containing 12-20 HDEHP monomers
- Ligand exchange kinetics studies of extractant phase lanthanide complexes indicate relatively labile $\text{Ln}(\text{DEHP}\cdot\text{HDEHP})_3$ ligand exchange reactions, but (on the NMR time scale) slow exchange/substitution-inert polynuclear complexes involving lactate
- Results of aqueous complexation kinetics studies suggest that the role of lactate as a phase transfer catalyst may arise from the transient presence of the electrically neutral $\text{Ln}(\text{Lac})_3$ complex on the aqueous side of the phase boundary
- Thermodynamic modeling of conventional TALSPEAK indicates that the extraction equilibrium should exhibit a slightly increasing dependence on pH rather than the often-observed strong negative dependence; though the system is difficult to model accurately due to the specific ion activity effects noted above, it is believed that competition by HLac for HDEHP reduces the effective extractant concentration which contributes to the pH dependence.
- Lactic acid extraction by $\text{HEH}[\text{EHP}]$ is significantly less than that of HDEHP; correspondingly, lanthanide extraction from a lactate/aminopolycarboxylate solution exhibits a pH dependence consistent with the predictions of a thermodynamic model, i.e., the strongly negative acid dependence of conventional TALSPEAK is eliminated
- Using $\text{HEH}[\text{EHP}]$ as a primary extractant for the Advanced TALSPEAK process requires the use of a matching weaker aminopolycarboxylate complexant; HEDTA and TTHA are viable actinide selective complexants in this application.

- Both citrate and lactate buffered acceptably in Advanced TALSPEAK
- Due to the reduced denticity of HEDTA, ternary complexes involving HEDTA and the buffer impact the separation efficiency of Advanced TALSPEAK
- Most of the research done on TALSPEAK has focused on 1,4-DIPB (emphasized here) and n-dodecane; the primary observation of Weaver and Kappelmann that aromatic diluents favor selectivity while aliphatic diluents enhance extraction must still be considered the definitive guidance regarding the choice of diluents for TALSPEAK.

With increased understanding of the fundamental interactions that drive and limit the utility of TALSPEAK, it would appear that recent advances should assist in efforts to improve the performance of conventional TALSPEAK applications. Perhaps more importantly, these new insights (and the natural evolution of progress in science and technology) have spurred extended applications of TALSPEAK-like chemistry to other separations platforms. One issue that has emerged from these extensions is that biphasic interactions increase with the number of component extractants, complexants, and buffers introduced. It would seem prudent to take advantage of modern analytical chemistry to probe the possible interactions between these species early-on in any process design exercise.

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TABLES

Table 1. Acid dissociation constants of DTPA in various ionic media.

Species	Ionic Strength					
	0.1 M ^a	0.5 M ^b	1.0 M ^c	2.0 M ^c	2.0 M ^c	2.0 M ^d
	NIST Database ¹⁷	Gritmon et. al. ³⁴	Tian and Rao ³⁵	Thakur et. al. ³⁶	Grimes ³¹	Grimes ³¹
	log K _a	log K _a	log K _a	log K _a	log K _a	log K _a
HL	10.48	9.86	9.43	9.46	9.50 ± 0.01	9.20 ± 0.01
H ₂ L	8.60	8.32	8.26	8.28	8.31 ± 0.01	8.06 ± 0.01
H ₃ L	4.28	4.12	4.23	4.13	4.38 ± 0.01	4.25 ± 0.01
H ₄ L	2.60	2.85	2.56	2.64	2.53 ± 0.03	2.68 ± 0.01
H ₅ L	2.00	1.95	2.37	2.12	2.41 ± 0.01	2.03 ± 0.03
H ₆ L	1.6					1.96 ± 0.02
H ₇ L	0.7					
H ₈ L	-0.1					

^aK⁺ salt used as background electrolyte.

^bKNO₃ used as background electrolyte.

^cNaClO₄ used as background electrolyte.

^dNaO₃SCF₃ used as background electrolyte.

Table 2. Lanthanide-DTPA stability constants $\log \beta_{101}$ and $\log K_{111}$ values for $\text{Ln}(\text{DTPA})^{2-}$ and $\text{Ln}(\text{HDTPA})^-$ complexes determined in 2.0 M NaClO_4 and NaSO_3CF_3 and in 0.1 M KNO_3 at 25 °C.

Metal	Ionic Strength		
	^a 0.1 M ³⁷ $\log \beta_{101}$	^a 2.0 M ³¹ $\log \beta_{101}$	^a 2.0 M ³¹ $\log \beta_{111}$
Y	22.05	20.13 ± 0.02	21.87 ± 0.04
La	19.48	18.02 ± 0.02	20.38 ± 0.03
Ce	20.50	19.06 ± 0.01	20.77 ± 0.02
Pr	21.07	19.64 ± 0.01	21.34 ± 0.02
Nd	21.60	20.23 ± 0.01	21.57 ± 0.05
Sm	22.34	20.79 ± 0.01	22.39 ± 0.02
Eu	22.39	21.03 ± 0.01	22.56 ± 0.03
Gd	22.46	21.15 ± 0.02	22.38 ± 0.02
Tb	22.71	21.15 ± 0.01	22.77 ± 0.02
Dy	22.82	21.23 ± 0.01	22.81 ± 0.02
Ho	22.78	21.43 ± 0.02	22.76 ± 0.05
Er	22.74	21.41 ± 0.02	22.66 ± 0.05
Tm	22.72	21.10 ± 0.02	22.45 ± 0.05
Yb	22.62	20.96 ± 0.01	22.53 ± 0.02
Lu	22.44	21.14 ± 0.01	22.21 ± 0.05
		2.0 M NaTriflate	
Eu		20.74 ± 0.01	22.48 ± 0.02

^a KNO_3 used as background electrolyte.

^b NaClO_4 used as background electrolyte.

TABLE 3 Recommended $\log \beta_{\text{lac}}$ and K_{ex} values (HDEHP extraction) for conventional TALSPEAK modeling calculations (I – 2.0 M, T = 25.0°C) (based on information found in references 17 and 33)

Lanthanide Lactate complex stability						Extraction Equilibria		
Ln	1/r cn8	$\log \beta_1$	$\log \beta_2$	$\log \beta_3$	ΔG_{MLac3}	$\log K_{\text{ex}}$	ΔG_{ex}	ΔG_{LnDTPA}
La	0.86199	2.27	3.95	5.1	-29.1	-2.22	+12.7	-102.8
Ce	0.87497	2.33	4.1	5.21	-29.7	-1.77	+10.1	-108.8
Pr	0.88802	(2.36)	(4.26)	(5.40)	-30.8	-1.46	+8.3	-112.1
Nd	0.90204	2.45	4.37	5.5	-31.4	-1.32	+7.5	-115.5
Pm	0.915	(2.45)	(4.49)	(5.67)	-32.4	-0.98	+5.6	-----
Sm	0.92704	2.56	4.58	5.9	-33.7	-0.50	+2.8	-118.7
Eu	0.938	2.53	4.6	5.88	-33.6	-0.22	+1.3	-120.0
Gd	0.95003	2.53	4.63	5.91	-33.7	-0.06	+0.3	-120.7
Tb	0.962	2.61	4.73	6.01	-34.3	+0.63	-3.6	-120.7
Dy	0.97399	2.71	4.76	6.25	-35.7	+0.95	-5.4	-121.2
Ho	0.98503	2.7	4.93	6.4	-36.5	+1.24	-7.1	-122.3
Er	0.99602	2.77	5.11	6.7	-38.2	+1.59	-9.1	-122.2
Tm	1.00604	(2.79)	(5.17)	(6.84)	-39.4	+1.99	-11.3	-120.4
Yb	1.01502	2.85	5.27	6.96	-39.7	+2.48	-14.2	-119.6
Lu	1.02396	(2.89)	(5.47)	(7.25)	-41.4	+2.75	-15.7	-120.6

Values in parentheses are interpolated based on Figure 3.

Table 4. Enthalpy values for the formation of $\text{Ln}\cdot\text{DTPA}^{2-}$ species in 0.1 M, 0.5 M, and 2.0 M ionic strength.

Metal	Ionic Strength		
	^a 0.1 M ³⁹	^b 0.5 M ³⁸	^c 2.0 M ³³
	$\Delta H /$ kJ/mol	$\Delta H /$ kJ/mol	$\Delta H /$ kJ/mol
Y	-27 ± 0.4	-36.3 ± 0.3	-35.0 ± 0.6
La	-22 ± 1.6	-25.3 ± 0.4	-30.3 ± 0.2
Ce	-24 ± 1.6	-29.9 ± 0.3	-31.2 ± 0.2
Pr	-26 ± 0.8	-35.5 ± 0.3	-35.5 ± 0.5
Nd	-30 ± 0.4	-38.4 ± 0.5	-38.0 ± 2.0
Sm	-33 ± 0.8	-44.2 ± 0.4	-41.0 ± 1.0
Eu	-32 ± 0.8	-47.8 ± 0.4	-42.6 ± 0.5
Gd	-31 ± 1.2	-47.5 ± 0.4	-42.0 ± 1.0
Tb	-31 ± 0.4	-44.1 ± 0.3	-42.2 ± 0.2
Dy	-31 ± 1.2	-44.7 ± 0.5	-42.6 ± 0.2
Ho	-30 ± 0.4	-42.6 ± 0.4	-41.0 ± 0.6
Er	-30 ± 0.8	-41.3 ± 0.4	-39.7 ± 0.4
Tm	-26 ± 0.4	-36.6 ± 0.3	-37.7 ± 0.9
Yb	-25 ± 0.8	-34.9 ± 0.4	-34.6 ± 0.3
Lu	-21 ± 0.4	-32 ± 0.5	-30.8 ± 0.4

^aK⁺ salt used as background electrolyte. ^bNaClO₄ used as background electrolyte. ^cNaNO₃ used as background electrolyte.

Table 5. Entropy values for the formation of Ln·DTPA²⁻ species in 0.1 M, 0.5 M, and 2.0 M ionic strength.

Ionic Strength				
	^a 0.1 M ³⁹	^b 0.5 M ³⁸	^c 2.0 M ³³	c 2.0 M
Metal	$\Delta S /$ J/mol·K	$\Delta S /$ J/mol·K	$\Delta S /$ J/mol·K	-T ΔS kJ/mol
Y	330	269 ± 2	268 ± 2	79.9 ± 0.6
La	298	264 ± 2	243 ± 1	72.4 ± 0.3
Ce	310	265 ± 2	260 ± 1	77.5 ± 0.3
Pr	314	257 ± 2	257 ± 2	76.6 ± 0.6
Nd	312	256 ± 2	261 ± 6	77.8 ± 1.8
Sm	315	248 ± 2	260 ± 5	77.5 ± 1.5
Eu	320	239 ± 2	260 ± 2	77.5 ± 0.6
Gd	322	238 ± 2	263 ± 3	78.4 ± 0.9
Tb	328	263 ± 2	263 ± 1	78.4 ± 0.3
Dy	332	254 ± 2	264 ± 1	78.7 ± 0.3
Ho	332	259 ± 2	273 ± 2	81.4 ± 0.6
Er	333	265 ± 2	277 ± 2	82.6 ± 0.6
Tm	345	284 ± 2	278 ± 3	82.9 ± 0.9
Yb	349	286 ± 2	285 ± 1	85.0 ± 0.3
Lu	357	290 ± 2	301 ± 1	89.7 ± 0.3

^aCalculated from ΔH and ΔG experiments where a K⁺ salt is used as background electrolyte.^b

Calculated from ΔH and ΔG experiments where NaClO₄ is used as background electrolyte.^c

Calculated from ΔH experiments where NaNO₃ is used as background electrolyte and ΔG experiments where NaClO₄ is used as background electrolyte.

Table 6. Calculated enthalpy of an ideal TALSPEAK extraction of europium. The calculated enthalpy for the entire process (net extraction enthalpy) is the sum of the enthalpies of all reactions known to be involved in the process (dissociation of EuDTPA^{2-} complex, extraction of Eu^{3+} to the organic layer, and reprotonation of DTPA^{5-}). All experimental values were determined at 2.0 M ionic strength and 25°C. A large discrepancy is noted between the calculated and experimental values.

Stage	Reaction	ΔH / kJ/mol
Breaking of EuDTPA^{2-} complex ³³	$\text{EuDTPA}^{2-} \rightleftharpoons \text{Eu}^{3+} + \text{DTPA}^{5-}$	42.6 ± 0.5
Extraction to organic layer (Zalupski et al. ²⁹)	$\text{Eu}^{3+} + 3(\text{HA})_{2,\text{org}} \rightleftharpoons \text{Eu}(\text{AHA})_{3,\text{org}} + 3\text{H}^+$	-13.1 ± 1.2
Reprotonation of DTPA^{5-} ³³	$\text{DTPA}^{5-} + \text{H}^+ \rightleftharpoons \text{HDTPA}^{4-}$	-30.3 ± 0.1
	$\text{HDTPA}^{4-} + \text{H}^+ \rightleftharpoons \text{H}_2\text{DTPA}^{3-}$	-24.6 ± 0.1
	$\text{H}_2\text{DTPA}^{3-} + \text{H}^+ \rightleftharpoons \text{H}_3\text{DTPA}^{2-}$	-9.8 ± 0.1
Calculated net extraction	$\text{EuDTPA}^{2-} + 3(\text{HA})_{2,\text{org}} \rightleftharpoons \text{Eu}(\text{AHA})_{3,\text{org}} + \text{H}_3\text{DTPA}^{2-}$	-35.2 ± 1.3
Experimental net extraction ¹⁸		-7.5

Table 7. Organic Phase Samples Prepared for SANS Analysis at $T = 22(\pm 1)^\circ\text{C}$ in p-Xylene- d_{10} .

sample [†]	extractant	Ln^{3+}	$[\text{HA,HB}]^\dagger$, M	$[\text{Ln}^{3+}]_{\text{org}}$, M	$[\text{HLac}]_{\text{org}}$, M	$[\text{H}_2\text{O}]_{\text{org}}$, M [‡]
HA _{0.1}	HDEHP	—	0.1004	—	—	—
HA _{0.2}	HDEHP	—	0.2002	—	—	—
HL _{0.1}	HDEHP	—	0.2002	—	0.0005	0.043
HL _{0.5}	HDEHP	—	0.2002	—	0.0032	0.043
HL _{1.0}	HDEHP	—	0.2002	—	0.0095	0.06
HL _{2.0}	HDEHP	—	0.2002	—	0.0408	0.124
HL _{1.0}	HEH[EHP]	—	0.1992	—	0.006	0.002
La _{0.003}	HDEHP	La ³⁺	0.1004	0.003	0.0048	—
Yb _{0.005}	HDEHP	Yb ³⁺	0.1004	0.005	0.0049	—
Gd _{0.013}	HDEHP	Gd ³⁺	0.1004	0.013	0.0127	—
La _{0.023}	HDEHP	La ³⁺	0.1004	0.023	0.02	—
La _{0.031}	HDEHP	La ³⁺	0.1004	0.031	0.0265	—

[†] HA = HDEHP, HB = HEH[EHP]

[‡] $[\text{H}_2\text{O}]_{\text{org}}$ for the Ln^{3+} metal systems was estimated to be 0.030 M for the scattering length density calculations.

Table 8. Summary of Guinier Analysis, Radius of Gyration, Weight Average Aggregation Number of HDEHP Extractant Samples in *p*-Xylene-d₁₀.

sample [†]	R _(spherical) , Å [§]	Mw, g/mol	n _w	n _w calc
HA _{0.1}	6.7 ± 0.5	620.4	1.92 ± 0.23	-
HA _{0.2}	6.7 ± 0.4	647.8	2.01 ± 0.23	-
HL _{0.1}	6.8 ± 0.1	647.1	2.01 ± 0.23	2.1
HL _{0.5}	7.1 ± 0.1	659.4	2.06 ± 0.24	2.05
HL _{1.0}	6.9 ± 0.1	668.6	2.08 ± 0.24	2.14
HL _{2.0}	7.9 ± 0.2	847.4	2.65 ± 0.31	2.62
HL _{1.0,HB}	6.2 ± 0.1	576.7	1.86 ± 0.22	-
La _{0.003}	8.0 ± 0.2	857.7	2.66 ± 0.32	2.65 [#]
Yb _{0.005}	7.8 ± 0.3	871	2.70 ± 0.33	2.71 [#]
Gd _{0.013}	8.3 ± 0.2	1130	3.50 ± 0.42	3.49 [#]
La _{0.023}	16.6 ± 0.4	4075.3	12.64 ± 1.50	macro
La _{0.031}	16.4 ± 0.5	5152.3	15.98 ± 1.52	macro

All uncertainties are reported at the 95% confidence level.

[†] HA = HDEHP, HB = HEH[EHP]

[§]R_{spherical} values calculated from R_g assuming spherical particles for HDEHP and HEH[EHP]aggregates.

[#] Calculated using equations 9-12, as explained in the text.

Table 9. Iterative fit of speciation results derived from Guinier analysis data for samples La₀₀₃, Yb₀₀₅, Gd₀₁₃ using mass balance relationships and expression defining the size of the average scattering particle considering Ln(DEHP·HDEHP)₃ (*Ln(DHD)*₃), Ln₂(Lac)₂(DEHP)₄ (*Ln₂L₂D₄*), HLac·(HDEHP)₃ (*HL(HD)*₃), and (HDEHP)₂ (*(HD)*₂) as the scattering particles.

sample	n _w (Guinier)	n _w (fit)	[Ln(DHD) ₃] M	[Ln ₂ L ₂ D ₄] M	[HL(HD) ₃] M	[(HD) ₂] M
La ₀₀₃	2.66 ± 0.32	2.65	0.0021	0.00045	0.0039	0.0743
Yb ₀₀₅	2.70 ± 0.33	2.71	0.00225	0.00138	0.00215	0.0750
Gd ₀₁₃	3.50 ± 0.42	3.49	0.00429	0.00436	0.00398	0.0453

Table 10. Complex formation (*k_f*) and dissociation (*k_d*) rate constants for lanthanide interactions with EDTA and DTPA in 0.3 M Na/H Lactate solutions at pH 3.6, I = 0.3 M (NaClO₄), [Ln-AAIII] = 5.0 x 10⁻⁵ M, [EDTA], [DTPA] = 1.0–6.0 x 10⁻⁴ M and 25.0°C. Values in parentheses represent ±1σ uncertainty limit.⁵⁷

Ln ³⁺	EDTA <i>k_f</i> (M ⁻¹ s ⁻¹) (x 10 ⁻³)	EDTA <i>k_d</i> (s ⁻¹)	DTPA <i>k_f</i> (M ⁻¹ s ⁻¹) (x 10 ⁻³)	DTPA <i>k_d</i> (s ⁻¹)
La ³⁺	15.9(0.2)	5.72(0.02)	28.2(1.0)	9.83(0.30)
Ce ³⁺	12.5(0.2)	4.92(0.20)	22.1(1.1)	8.99(0.36)
Pr ³⁺	16.4(0.5)	4.46(0.06)	22.4(0.8)	5.88(0.22)
Nd ³⁺	15.8(0.4)	3.52(0.16)	21.9(1.6)	4.06(0.22)
Sm ³⁺	17.7(0.6)	6.45(0.14)	23.4(1.4)	2.82(0.40)
Eu ³⁺	20.9(0.7)	6.76(0.15)	21.0(1.7)	4.23(0.23)
Gd ³⁺	33.0(1.6)	6.10(0.40)	30.5(2.0)	1.69(0.56)
Tb ³⁺	27.0(1.2)	3.31(0.24)	22.4(0.9)	1.61(0.24)
Dy ³⁺	22.9(1.1)	0.75(0.31)	24.1(0.7)	0.77(0.18)
Ho ³⁺	22.7(0.9)	0.58(0.25)	21.6(0.2)	-
Er ³⁺	19.6(0.8)	0.51(0.25)	19.1(0.9)	-
Tm ³⁺	12.0(0.7)	1.07(0.10)	17.1(0.7)	-
Yb ³⁺	8.9 (0.8)	0.99(0.23)	14.2(0.9)	-
Lu ³⁺	8.3 (0.7)	0.78(0.20)	8.9(0.1)	0.44(0.02)

FIGURES

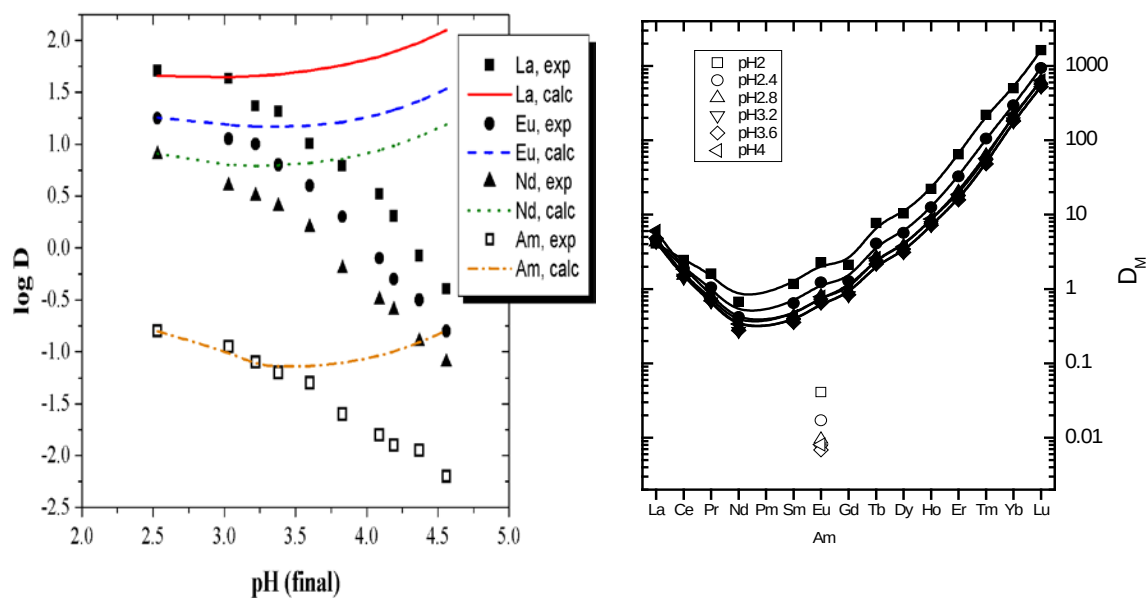


Figure 1. Calculated vs. observed partitioning in conventional TALSPEAK. Adapted from reference 17. (left), Calculated lanthanide and Am^{3+} partitioning on conventional TALSPEAK ($T=25^\circ\text{C}$, $I = 2.0 \text{ M}$, 1.0 M Lactate, 0.05 M DTPA, 0.2 M HDEHP).

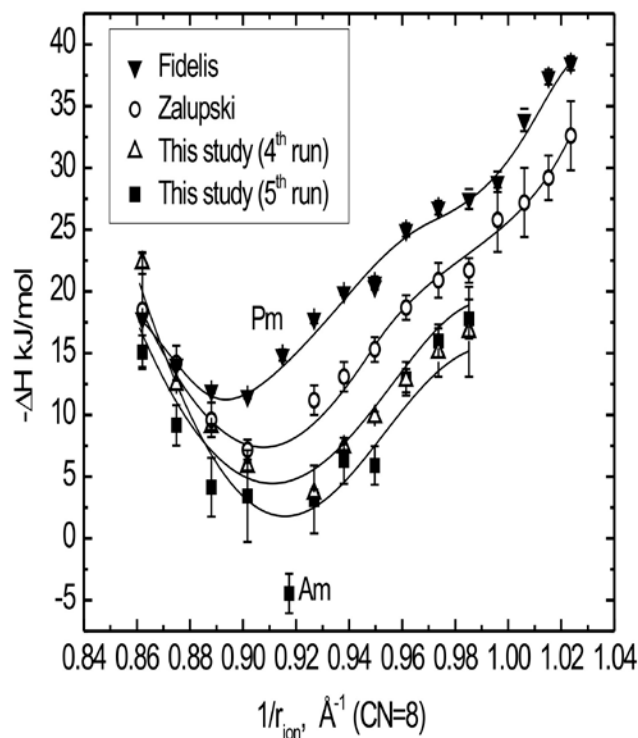


Figure 2. Conditional enthalpies (based on van't Hoff analysis of D_m for extraction of lanthanides and Am^{3+}).¹⁸ Aqueous medium 1 M lactic acid, 0.05 M DTPA, 1 mM total Ln + Y, (4th run) 1 M total nitrate, p[H] 3.7, (5th run) 1 M total sodium, p[H] 4.1. Organic medium: 1,4-DIPB containing 0.5 M HDEHP (4th run) or 0.2 M HDEHP (5th run). Results from the literature on the extraction of lanthanides from mineral acid solutions are for HDEHP in n-heptane^{27, 28}, n-dodecane²⁹

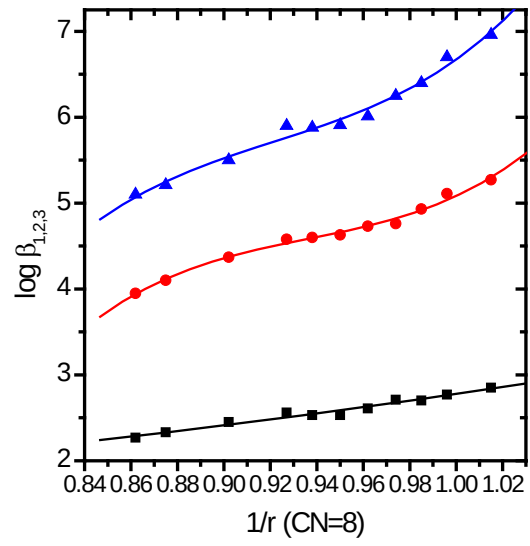


Figure 3. Correlation of experimental data on lanthanide-lactate complexes from NIST¹⁷ and estimates of missing/inconsistent data describing lanthanide lactate stability constants $I = 2.0\text{ M}$, $T = 25.0^\circ\text{C}$

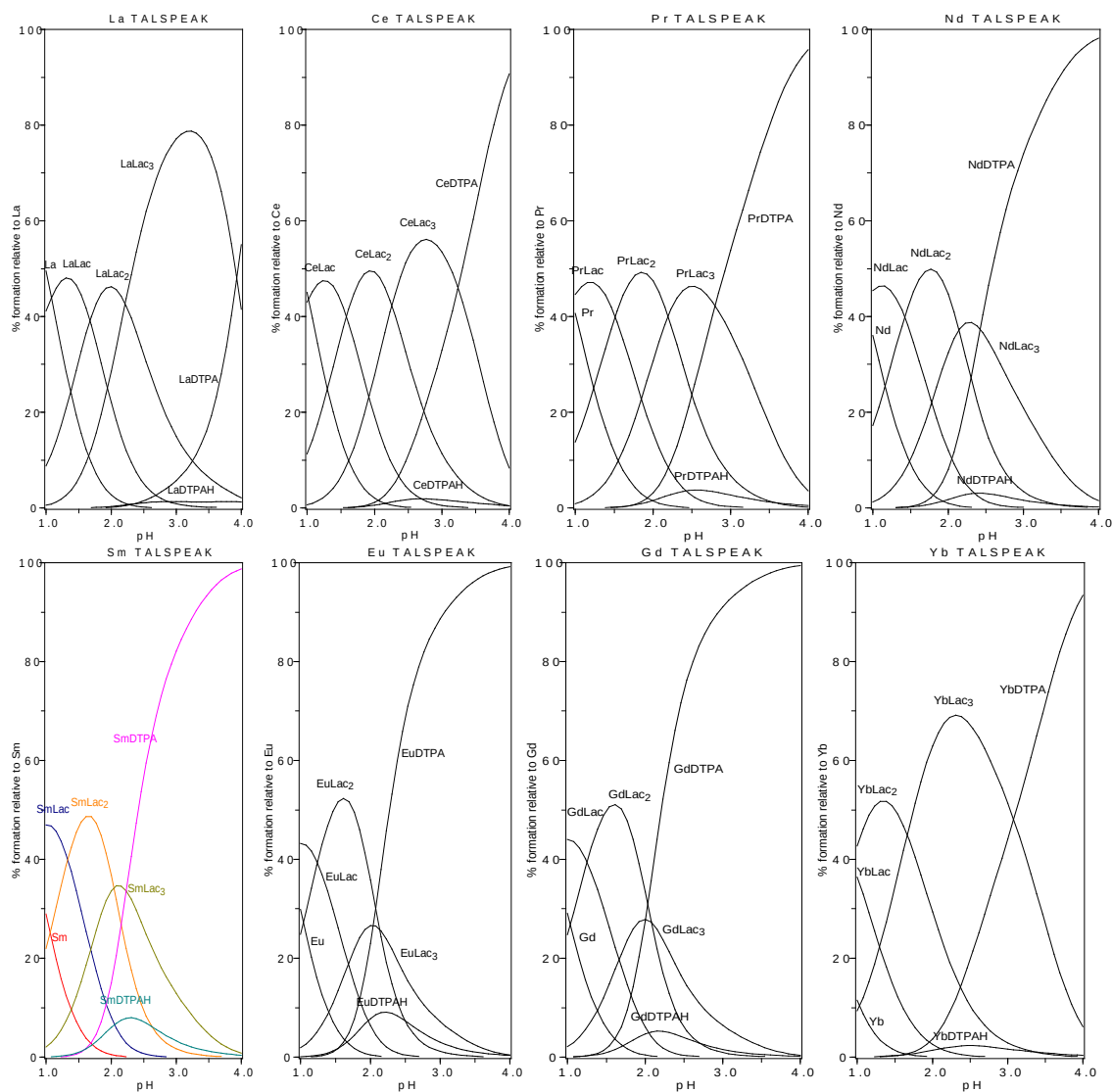


Figure 4. Predicted stability field diagrams for selected lanthanide DTPA and lactate complexes under TALSPEAK-relevant conditions ($[\text{DTPA}]_t = 0.01\text{ M}$, $[\text{Na/HLac}] = 2.0\text{ M}$, $T = 25^\circ\text{C}$, $I = 2.0\text{ M}$, $[\text{Ln}^{3+}] = 0.001\text{ M}$) (Adapted from reference 31)

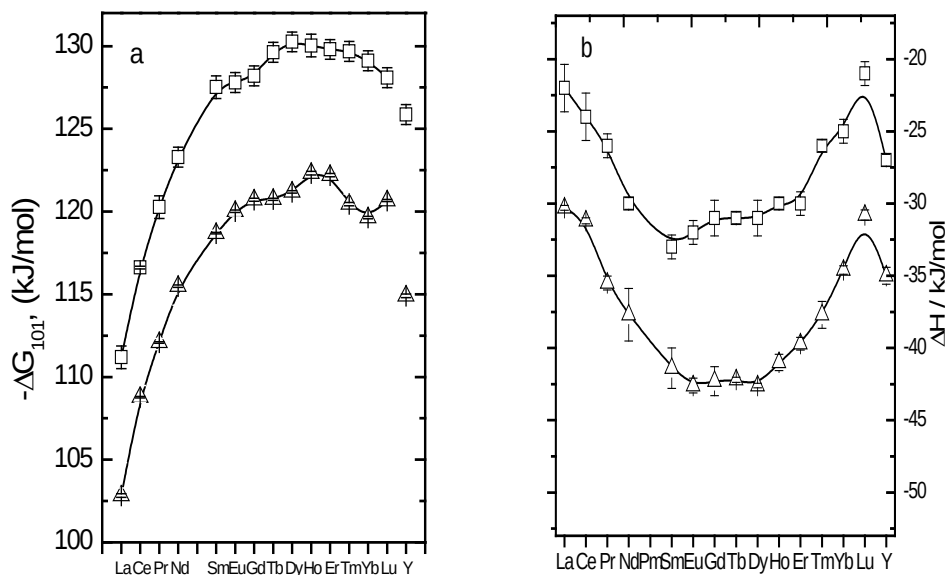


Figure 5 a. Available free energy data for the Ln·DTPA²⁻ complex from the literature at different ionic strengths ($\square$) 0.1 M ³⁴, (Δ) 2.0 M ³³ adjusted with NaClO₄, **b.** Available enthalpy data for the Ln·DTPA²⁻ complex from the literature at different ionic strengths ($\square$) 0.1 M (K⁺ salt) ³⁹, (Δ) 2.0 M (NaNO₃, ³³) .

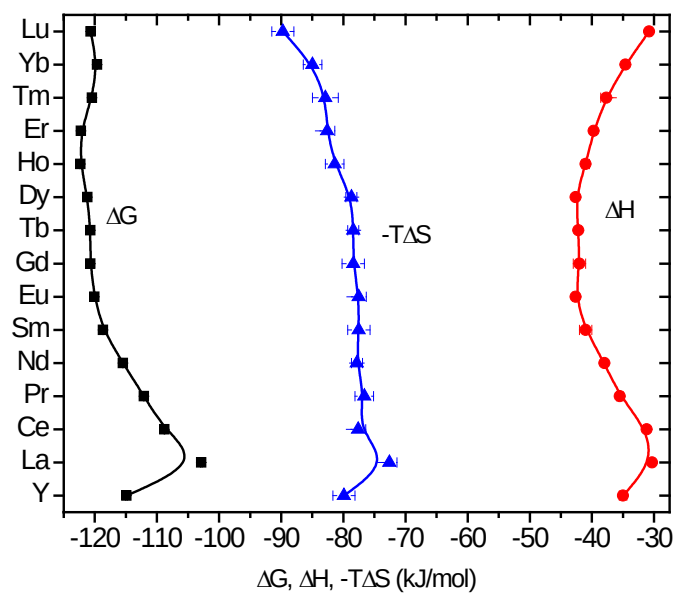


Figure 6. Free energy, enthalpy, and entropy describing the formation of lanthanide tris-lactate complexes at 25°C, I = 2.0 M.¹⁷

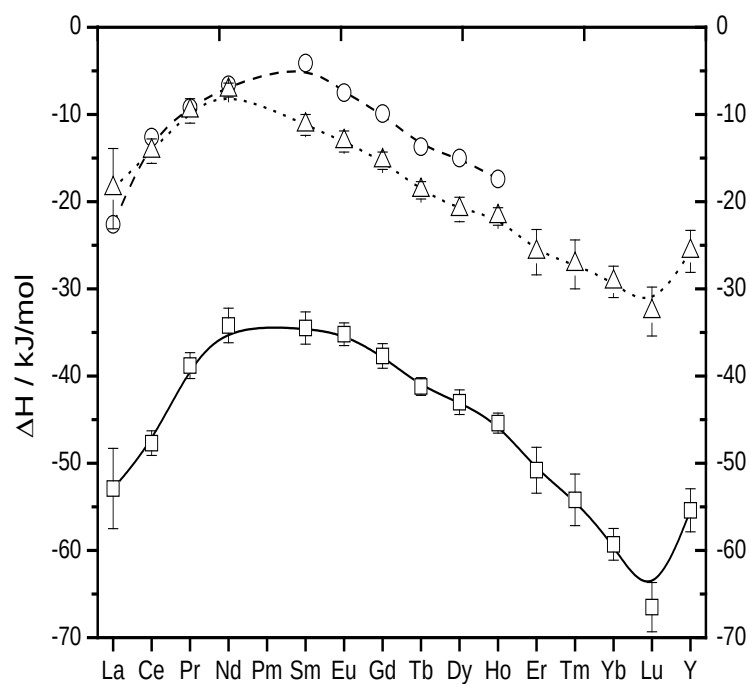


Figure 7. Enthalpies of: the extraction of the lanthanides by HDEHP alone (Δ)²⁹; the ideal TALSPEAK system as modeled by sum of the individual enthalpies of the extraction reaction, the M-DTPA complexation reaction, and all relevant protonation steps ($\square$); and the entire TALSPEAK system, as experimentally determined ($\circ$)¹⁸. All values are at $I = 2.0$ M and 25°C. Lines are included to guide the eye, and do not denote any mathematical fit.

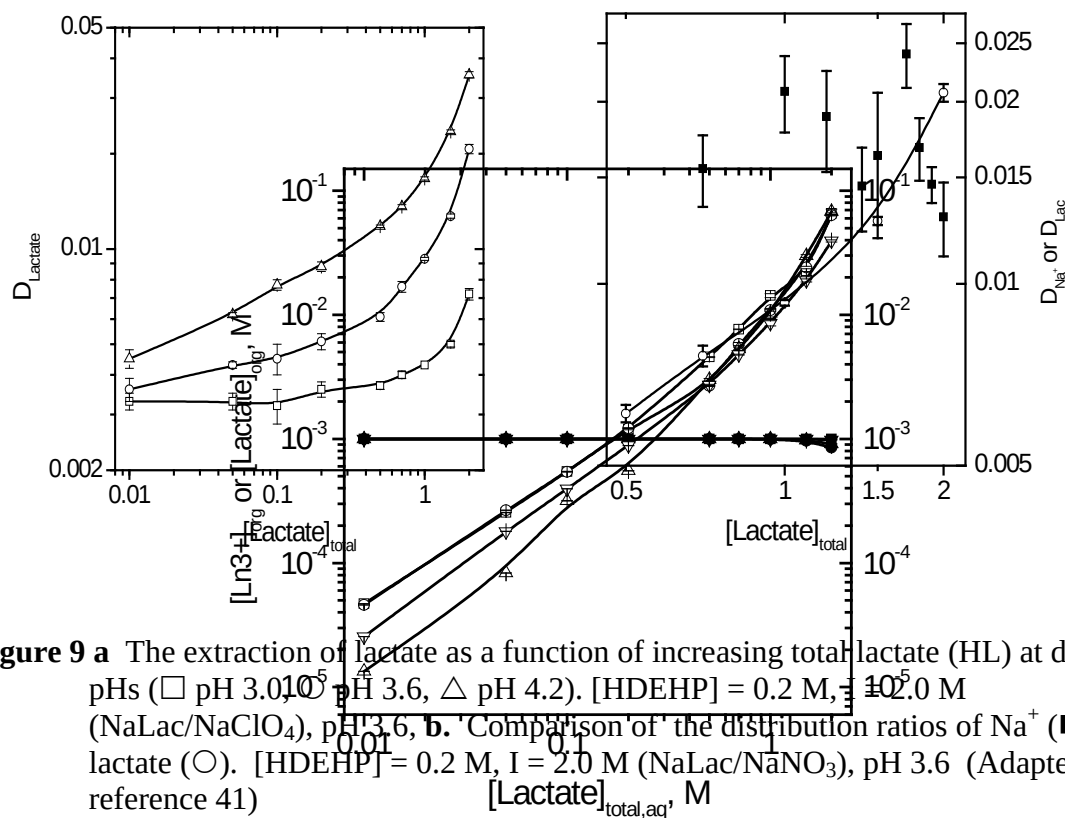


Figure 9 a The extraction of lactate as a function of increasing total lactate (HL) at different pHs ($\square$ pH 3.0, $\circ$ pH 3.6, $\triangle$ pH 4.2). $[\text{HDEHP}] = 0.2 \text{ M}$, $[\text{NaLac}/\text{NaClO}_4] = 10^{-5} \text{ M}$, pH 3.6. b. Comparison of the distribution ratios of Na^+ ($\blacksquare$) and lactate ($\circ$). $[\text{HDEHP}] = 0.2 \text{ M}$, $I = 2.0 \text{ M}$ ($\text{NaLac}/\text{NaNO}_3$), pH 3.6 (Adapted from reference 41)

Figure 8. Concentration of the Ln^{3+} and lactate in the organic phase at equilibrium. The metals are represented by the following symbols $\blacksquare \text{Y}^{3+}$, $\bullet \text{La}^{3+}$, $\blacktriangle \text{Nd}^{3+}$, $\blacktriangledown \text{Eu}^{3+}$, the lactate and the accompanying metal system are denoted with $\square \text{lac}_{\text{Y}}^{3+}$, $\circ \text{lac}_{\text{La}}^{3+}$, $\triangle \text{lac}_{\text{Nd}}^{3+}$, $\nabla \text{lac}_{\text{Eu}}^{3+}$. $[\text{HDEHP}] = 0.2 \text{ M}$, $I = 2.0 \text{ M}$ ($\text{NaLac}/\text{NaNO}_3$), pH 3.6 (Adapted from reference 41)

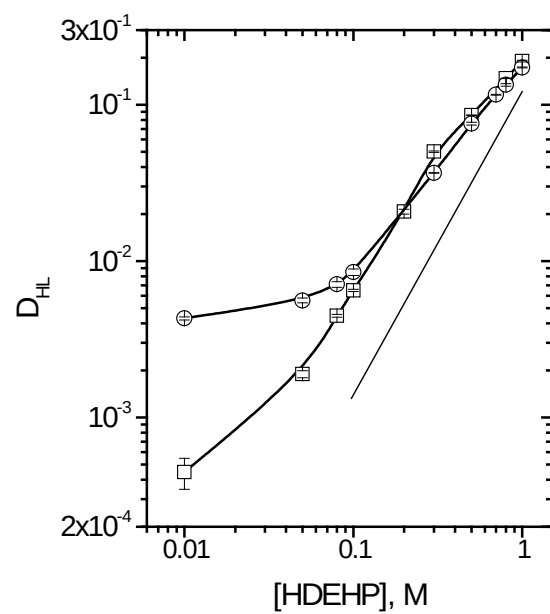


Figure 10. Comparison of the extraction behavior of HL at 1 M (○) and 2 M (□) lactate as a function of HDEHP concentration in 1,4-di-isopropylbenzene, pH = 3.6, I = 2.0 M (NaLac/NaNO₃). Adapted from reference 41

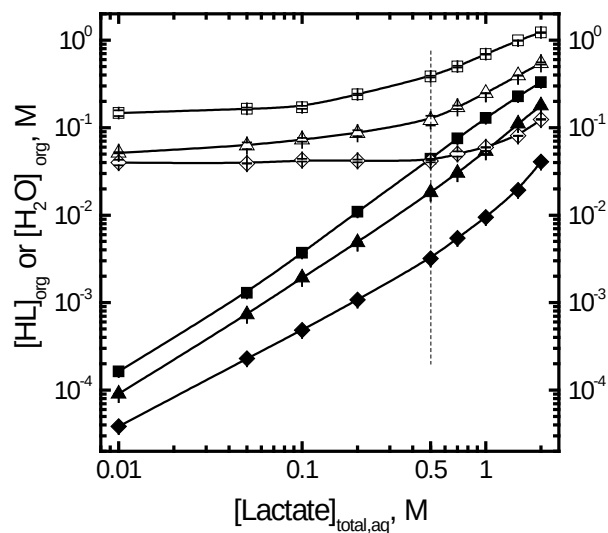


Figure 11. Comparison of organic phase H₂O concentration (open symbols) compared to HLac concentrations (closed symbols) at different HDEHP concentrations. (■HLac, □H₂O) 1.0 M HDEHP, (▲HLac, △H₂O) 0.5 M HDEHP, (◆HLac, ◇H₂O) 0.2 M HDEHP (0.2 M HDEHP data previously reported in ref. 11). Water content in the organic phase was determined by Karl Fischer titration. The trend lines in the graph are only to guide the eye.

2.0 M Na/NO₃⁻Lac⁻ / 0.5 M HDEHP in 1,4-diisopropyl benzene

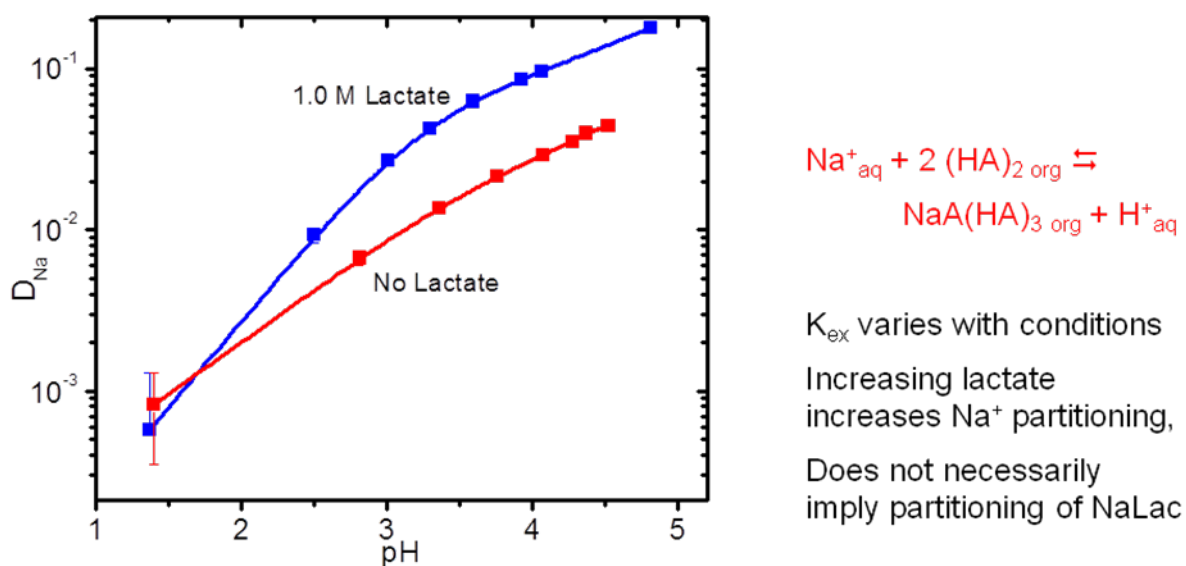


Figure 12. Partitioning of Na⁺ into 0.5 M HDEHP/1,4-DIPB from 2.0 M NaNO₃ and from 1.5 M NaNO₃/ 1.0 M NaLac at pH 3.6.⁴⁴

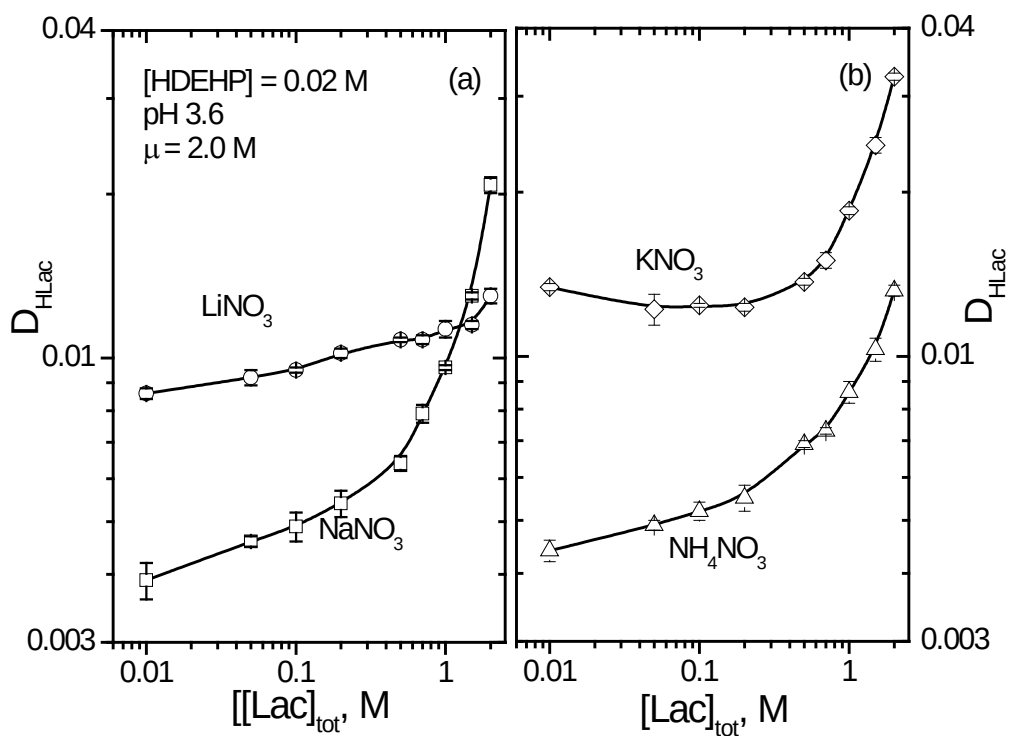


Figure 13. a) Distribution of lactic acid into 0.2 M HDEHP in 1,4-diisopropylbenzene from (○) LiNO_3 and (□) NaNO_3 . b) Distribution of lactic acid from (◇) KNO_3 and (△) NH_4NO_3 . Aqueous phases were 0-2 M lactic acid, constant 2 M ionic strength (adjusted with MNO_3) and a fixed pH 3.6. Adapted from reference 41

[Lac ⁻] _{tot,aq} M	0.2 M HDEHP	0.5 M HDEHP	1.0 M HDEHP
	k _{obs} (s ⁻¹)	k _{obs} (s ⁻¹)	k _{obs} (s ⁻¹)
0	339 ± 10	364 ± 9	382 ± 8
0.05	338 ± 9	359 ± 10	376 ± 9
0.1	337 ± 7	-	-
0.5	347 ± 11	359 ± 9	378 ± 9
1.0	345 ± 12	-	-
1.5	355 ± 10	-	-
2.0	367 ± 11	389 ± 10	402 ± 7

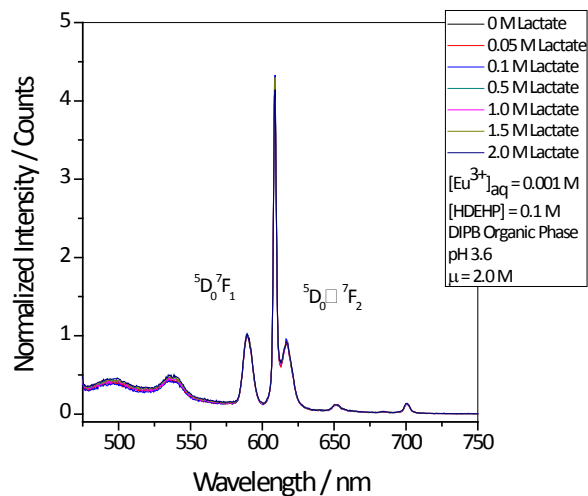


Figure 14. Emission spectra for the extracted Eu³⁺ complexes of 0.1 M HDEHP and varying aqueous phase lactate concentrations. (Adapted from Reference 49).

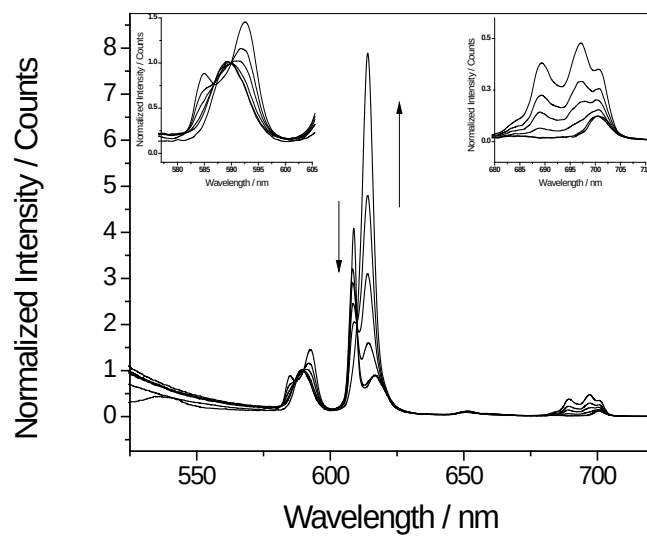


Figure 15. Emission spectra for extracted Eu³⁺ complexes under loading conditions. La³⁺ was used to load the organic phase with a 0.001 M Eu³⁺ spike to promote fluorescence. Aqueous lactate concentration was 1.0 M, pH 3.6, I = 2.0 M (NaNO₃). The La³⁺ concentrations for metal loading were, 0.001 M, 0.003 M, 0.010 M, 0.017 M, 0.025 M, and 0.037 M. (adapted from reference 49)

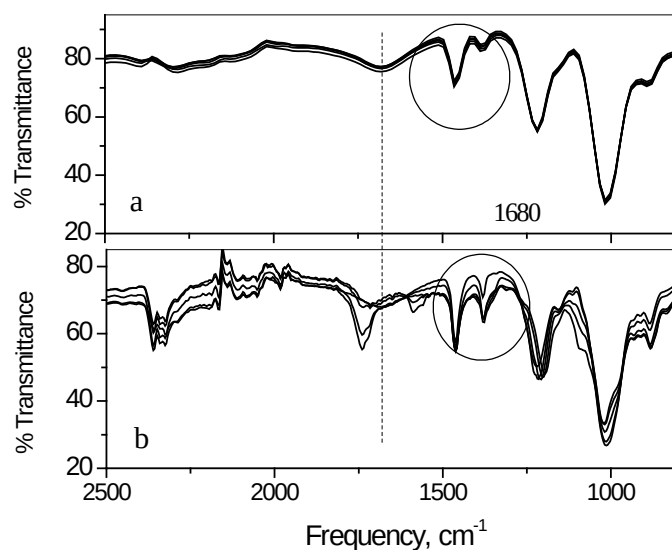


Figure 16. IR spectra for evaporated residues of HDEHP organic phases loaded with $[La^{3+}] = 0.0\text{ M}, 0.003\text{ M}, 0.010\text{ M}, 0.017\text{ M}, 0.025\text{ M}$ La^{3+} . Panel a shows the effect of metal loading (same as lactate systems) with no lactate present in the aqueous phase. Nitric acid was used to adjust the acidity to a pH of 3.0. $I = 2.0\text{ M}$ for both extraction systems. Panel b shows the effect of metal loading with lactate buffer present in the aqueous phase at 1.0 M. The acidity of the aqueous phase was adjusted to a pH of 3.6. (adapted from reference 49)

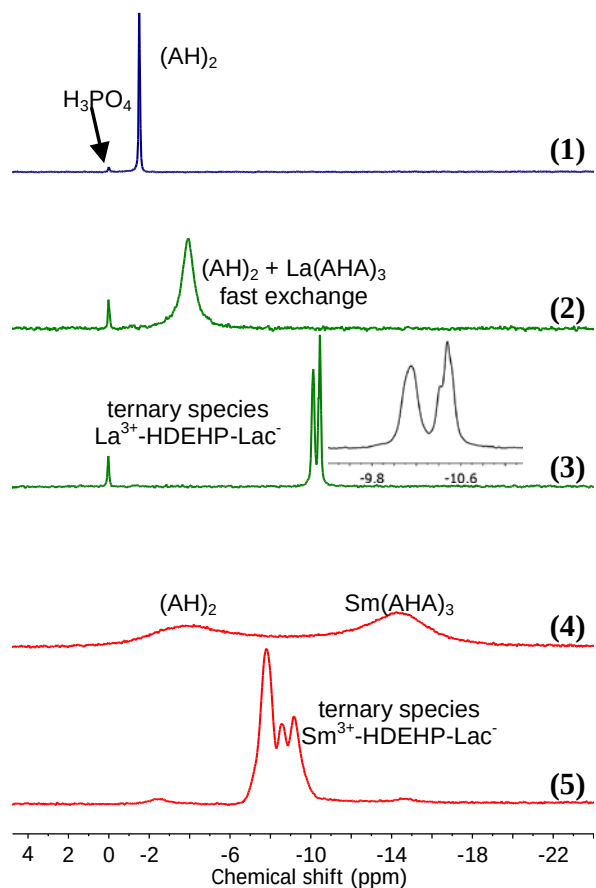


Figure 17. ^{31}P NMR spectra of HDEHP solutions in 1,3-dipb at 22°C; **(1)** HDEHP 0.1 mol/L in 1,3-dipb; **(2)** HDEHP 0.1 mol/L after equilibration with $\text{La}(\text{NO}_3)_3$ 86 mmol/L at pH=3; **(3)** HDEHP 0.1 mol/L after equilibration with $\text{La}(\text{NO}_3)_3$ 55 mmol/L at pH=3.6, $[\text{lactate}]_{\text{tot.}} = 0.5$ mol/L, $[\text{NO}_3^-] = 1.5$ mol/L; **(4)** HDEHP 0.1 mol/L after equilibration with $\text{Sm}(\text{NO}_3)_3$ 0.112 mol/L at pH=3; **(5)** HDEHP 0.1 mol/L after equilibration with $\text{Sm}(\text{NO}_3)_3$ 54 mmol/L at pH=3.4, $[\text{lactate}]_{\text{tot.}} = 0.5$ mol/L, $[\text{NO}_3^-] = 1.5$ mol/L; H_3PO_4 in *d*-acetone used as chemical shift reference ($\delta = 0.00$ ppm) for spectra (1), (2) and (3) and $\text{Ph}_3\text{P}=\text{O}$ in *d*-DMSO ($\delta = 23.23$ ppm) for spectra (4) and (5); $(\text{AH})_2$ = dimeric HDEHP; AHA^- = singly deprotonated HDEHP dimer (Adapted from reference 50)

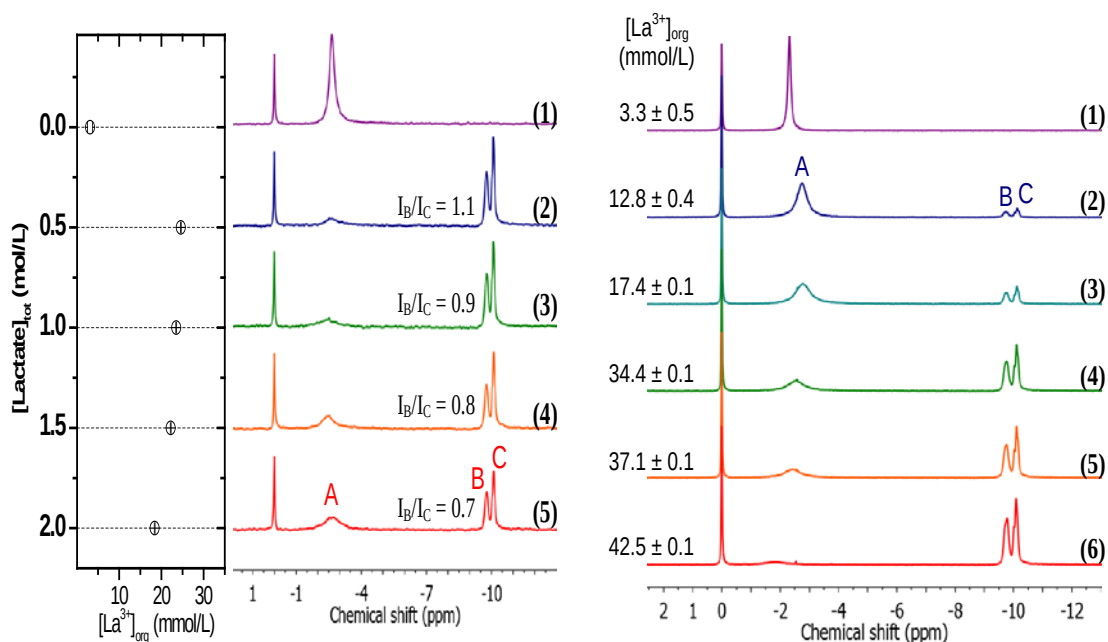


Figure 18 *left* ^{31}P NMR spectra ($T = 22^\circ\text{C}$) of HDEHP 0.1 mol/L in 1,3-dipb after equilibrium with an aqueous phase containing $\text{La}(\text{NO}_3)_3$ 25 mmol/L at 2 mol/L of ionic strength (adjusted with NaNO_3) and an increasing concentration of lactate: (1) no lactate, (2) $[\text{Lactate}]_{\text{tot}} = 0.5$ mol/L, (3) $[\text{Lactate}]_{\text{tot}} = 1.0$ mol/L, (4) $[\text{Lactate}]_{\text{tot}} = 1.5$ mol/L, (5) $[\text{Lactate}]_{\text{tot}} = 2.0$ mol/L; H_3PO_4 in d-acetone was used as chemical shift reference ($\delta = 0$ ppm); *right* ^{31}P NMR spectra ($T = 22^\circ\text{C}$) of HDEHP 0.1 mol/L in 1,3-dipb after equilibrium with an aqueous phase at pH = 3.6 containing $\text{La}(\text{NO}_3)_3$ at various concentrations; $[\text{lactate}]_{\text{tot}} = [\text{NaNO}_3] = 1$ mol/L. On the left side of the figure are reported the concentrations of lanthanum in the organic phase after equilibrium. H_3PO_4 in d-acetone was used as chemical shift reference ($\delta = 0$ ppm). (Adapted from information in reference 51)

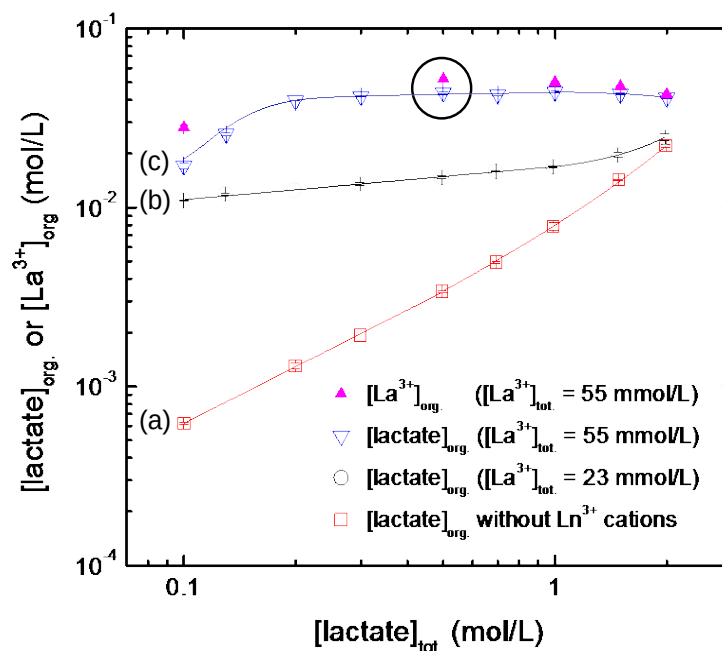


Figure 19. Concentrations of lactate and La^{3+} extracted by HDEHP at 0.1 mol/L in 1,3-dipb as a function of the total concentration of lactic acid. Aqueous phases were at pH = 3.6 with trace amounts of ^{14}C lactic acid, lanthanum nitrate (23 and 55 mmol/L) and constant 2 mol/L ionic strength (adjusted with NaNO_3). (Adapted from reference 51)

Figure 20. (a) Structure of $\text{La}(\text{AHA})_3$ binary complex, (b) and (c) possible structures of the ternary complexes $(\text{LaA}_2\text{Lac})_n$ with $n = 1$ and 2

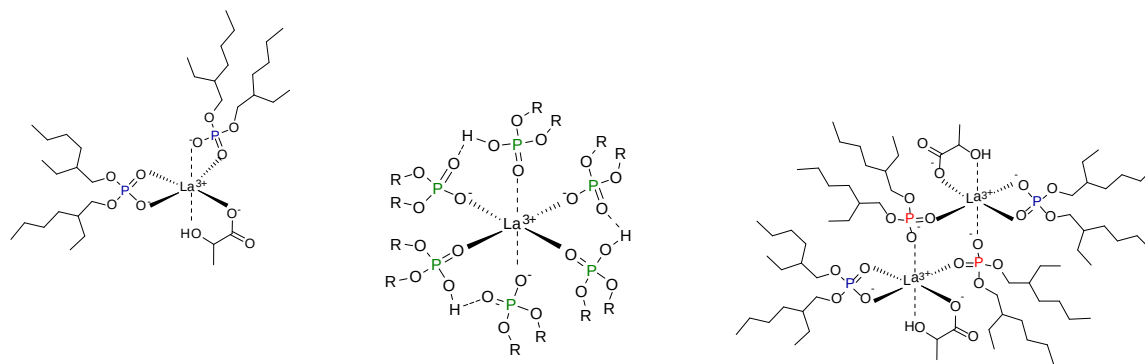
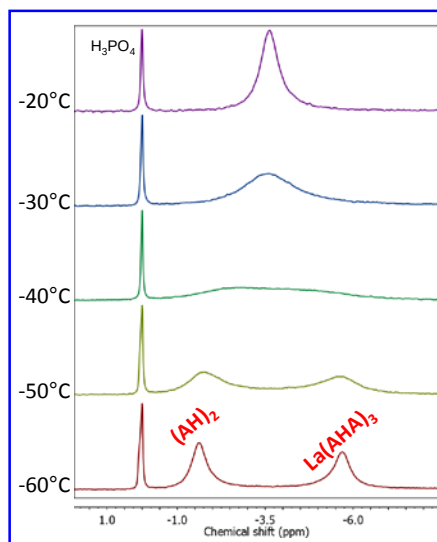


Figure 21. Experimental conditions, rates and activation parameters for the HDEHP exchange reaction in $\text{Ln}(\text{DEHP} \cdot \text{HDEHP})_3$ complexes ($\text{Ln} = \text{La}$ and Sm) at 25°C in 1,3-DIPB. Quoted errors correspond to standard deviations obtained from the linear regression fit of experimental data to the Arrhenius equation.



Organic phase:

0.1 M HDEHP in 1,3 DIPB

Aqueous phase:

0.1 M $\text{La}(\text{NO}_3)_3$, pH = 3

$[\text{La}^{3+}] = 0.0089 \text{ M}$

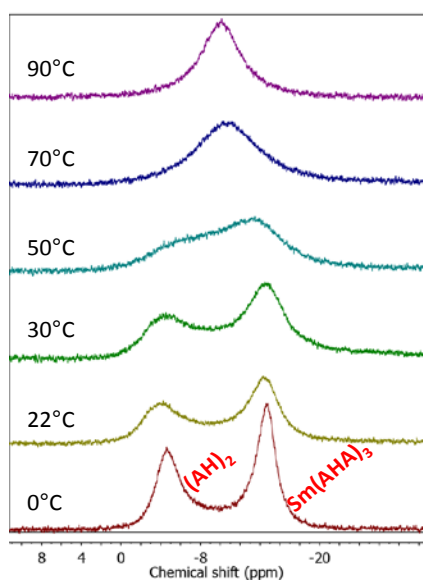
%HDEHP Complexed = $53.5 (\pm 1.1)$

HDEHP/ La^{3+} in complex = $5.8 (\pm 0.1)$

$$k_1 = 1.3 \times 10^5 \text{ s}^{-1} (25.0^\circ\text{C})$$

$$\Delta H^\ddagger = 48.3 \pm 1.6 \text{ kJ/mol}$$

$$\Delta S^\ddagger = 15 \pm 6 \text{ J/mol-K}$$



Organic phase:

0.1 M HDEHP in 1,3 DIPB

Aqueous phase:

0.1 M $\text{Sm}(\text{NO}_3)_3$, pH = 3

$[\text{Sm}^{3+}] = 0.0102 \text{ M}$

%HDEHP Complexed = $61.0 (\pm 0.5)\%$

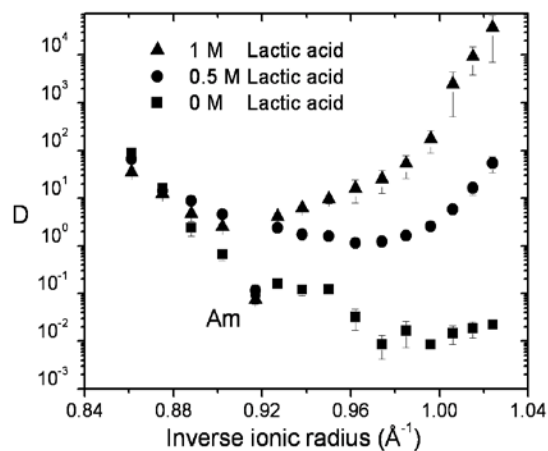
HDEHP/ Sm^{3+} in complex = $5.7 (\pm 0.2)$

$$k_1 = 1.8 \times 10^3 \text{ s}^{-1} (25.0^\circ\text{C})$$

$$\Delta H^\ddagger = 17.0 \pm 0.9 \text{ kJ/mol}$$

$$\Delta S^\ddagger = -126 \pm 3 \text{ J/mol-K}$$

Figure 22. Results of extraction of Ln and Am under TALSPEAK conditions: 0.3 M HDEHP in n-dodecane, 0.05 M DTPA, 1 mM Ln^{3+} , ^{241}Am , 1.0 M Na^+ , pH 3.5, Contact time (vortex mixing) 30 min. (Adapted from Reference 54).



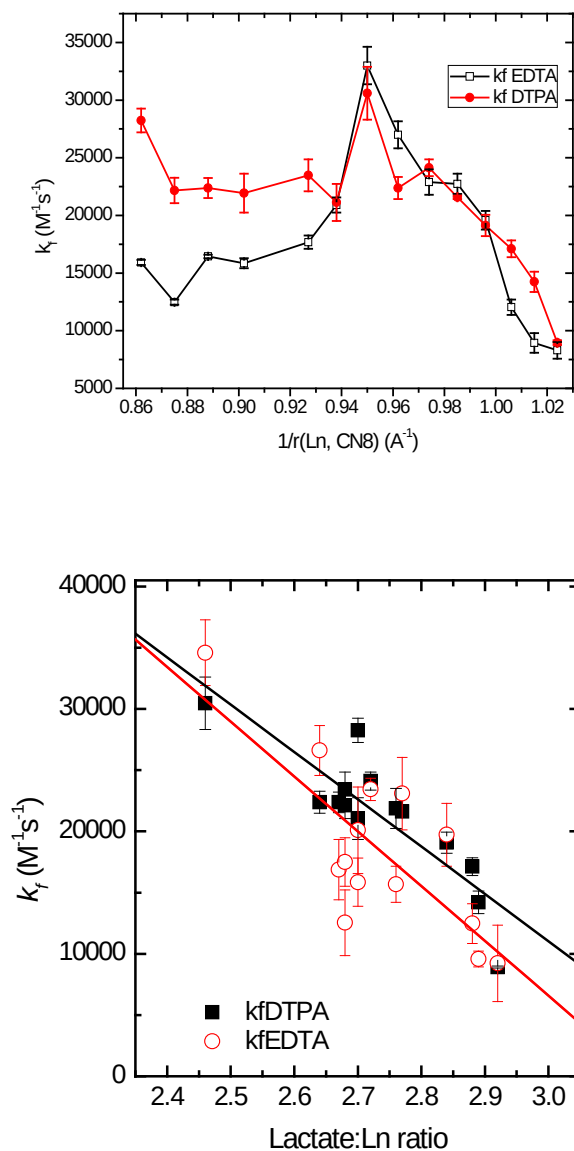


Figure 23 top: Correlation of complex formation rate constants across the Ln series with inverse cation radius at pH 3.6, 25 °C, and $[\text{Lac}_{\text{tot}}] = 0.3 \text{ M}$ (EDTA ($\square$), DTPA, **bottom:** Rate constant for formation of lanthanide EDTA and DTPA complexes in 0.3 M sodium lactate buffer, pH = 3.6, $T = 25.0^\circ\text{C}$, $I = 0.3 \text{ M}$ as a function of the average equilibrium Lactate:Ln complex stoichiometry ($\bar{n}_{\text{LacLn}}$).

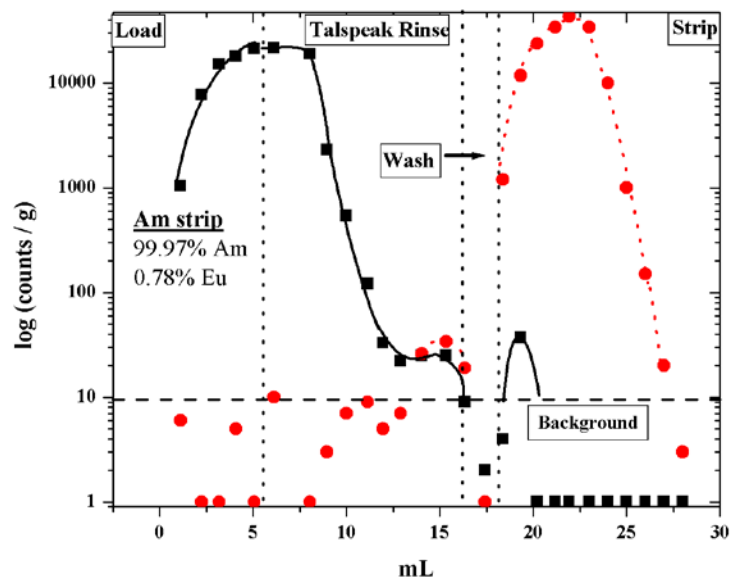


FIGURE 24. Optimized Am(III), Eu(III) elution curve separation LN2 S-grade (50-100 μm) Amberchrom GC71 resin. **Flow rate:** 1 mL / min. Temperature: 50° C. **Load:** 5 mM TTHA, 1 M NO_3^- , pH 3.6, 1 mM Eu, Trace Am TALSPEAK **Rinse:** 5 mM TTHA, 1 M NO_3^- , pH 3.6 **Wash:** 0.001 M HNO_3 **Strip:** 1 M HNO_3 . (Adapted from reference 64)

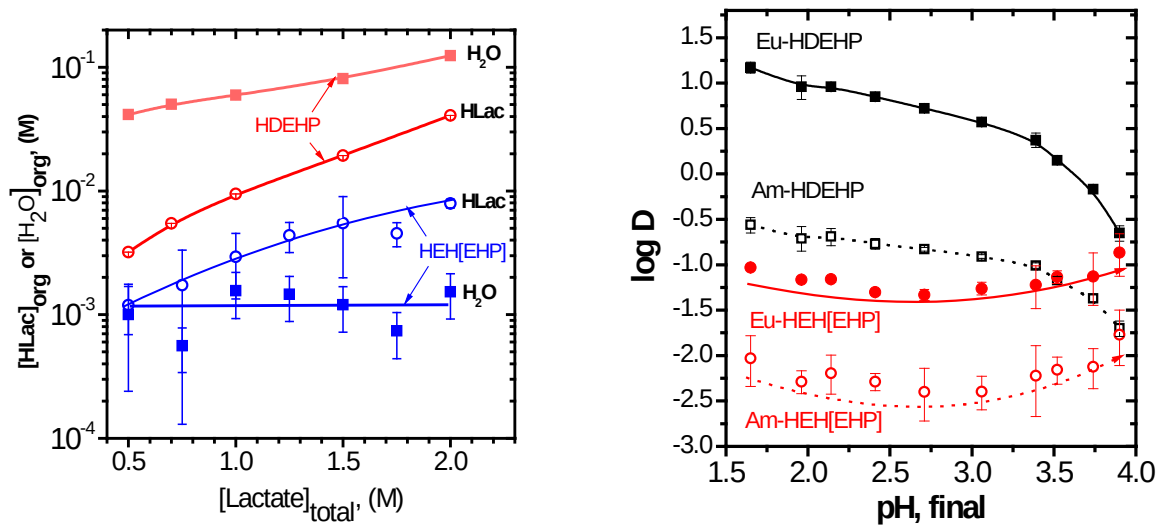


FIGURE 25 Left. Comparison of water and lactate partitioning into HDEHP or HEH[EHP]/n-dodecane, Right. Comparison of Am^{3+} and Eu^{3+} partitioning in Conventional TALSPEAK (HDEHP) and Advanced TALSPEAK (HEH[EHP]) solvents

Organic: 0.1 M HDEHP or HEH[EHP]

Aqueous: 20 mM DTPA, 1.0 M nitrate, 1.0 M total lactic acid

*1 σ error indicated

Adapted from reference 67.

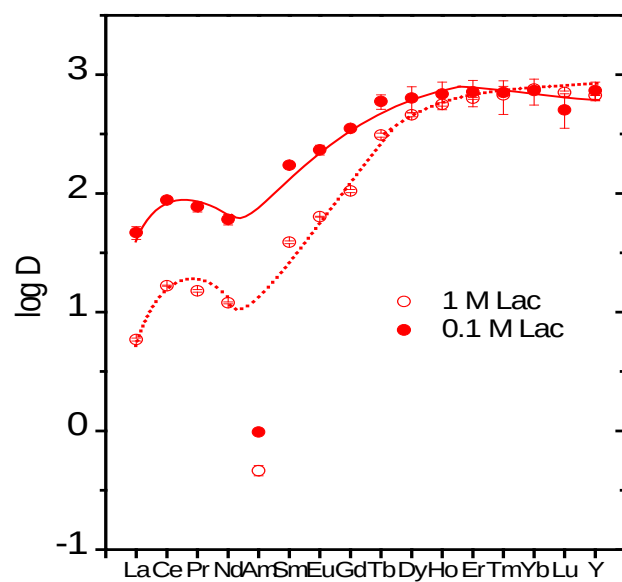


FIGURE 26 Trans-lanthanide extraction pattern in Advanced TALSPEAK, 0.1 M HEH[EHP]/n-dodecane, 0.1 or 1.0 M Lactate, HEDTA. (Adapted from reference 67)